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SAMSUNG SBB-CS4B INSTALLATION MANUAL



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23Y SBox Installation Manual

Model Name: SBB-CS4B

LED Signage Box for Indoor Cabinet Series

Model Code: SBB-CS4BPGS

Ver 1.6

Revision History

Version	Date (Y/M/D)	Description
1.0	2023/03/22	New Release
1.1	2023/04/04	Add Multi -Link HDR
1.2	2023/04/13	Add Eco sensor contents
1.3	2023/05/25	LSM specification update
1.4	2023/06/14	Add IG power case
1.5	2023/08/29	Add Device Care
1.6	2023/11/20	Add Appendix

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4. Setting andguide

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1.Product’s Description

1. Product's Description

❑ This manual is the installation manual of CS4B model, which is S-BOX model exclusively for Indoor LED.

Model	Product Appearance & Information
SBBCS4BPGS/**	
Remarks	☐ Main features - Switch to JMU and increase compatibility of fine pitch cabinets ・4K support Affordable price Box secured - Performance improvement compared to JMU ・Apply new SoC image quality and Smart function (AI processing, Smartview+, Device Care) ・ 4K 60Hz 4:4:4 10bit support by applying HDMI 2.1 / DP 1.4 ・Application of MultiLink HDR function: Technology to optimize dynamic picking operation between multiple S-BOXs as one screen

1. Product's Description

❑ Model Specification

		CS4B (Standard)	CV4B (VP)	CX4B (Outdoor)
Support Model		IWA-VP(P1.68), IFR/IEA, IWC	IWA-VP(P1.68), IFR/IEA, IVC	XHB, XLB
Maximum resolutions		4K UHD (3840 x 2160)		
Frame Rate		Up to 60Hz		
Input	**DP 1.4a**	1	1	1
	HDMI 2.1	2	2	2
	USB 2.0	2	2	2
	12G-SDI	-	1	-
Output	**Optic (QSFP+)**	4 (SR4, LR4)	4 (SR4, LR4)	4 (LR4, SR4)
Others		LAN	LAN	LAN
		Service port (DVI type)	Service port (DVI type)	Service port (DVI type)
		RS232C In	RS232C In	RS232C In
		Digital Audio Out (Optical)	Digital Audio Out (Optical)	-
Function		AI Processing, HDR10, HDR10+, Multi Link HDR, LED HDR	AI Processing, HDR10, HDR10+	
		Gen-lock, Frame-lock, Phase Offset		
		Redundancy, Multi-Box		
		MagicINFO (CMS/Diagnotics, RM)		
		DICOM Simulation Mode, Director Mode	-	-
Standard		EMC Class B	EMC Class B	EMC Class A
Solution		Remote diagnosis	Remote diagnosis, VP Solution	Remote diagnosis, Dimming

	I/G	Z-Card (Separate sale)
Connectivity	Data In / Data Out	Ethernet
	IR	-
	Eco Sensor ※ CV4B - Eco Sensor Unsupported	Light Sensor

1. Product's Description

❑ Model Specification - Product comparison

ITEM		SBB-CS4B	SBB-SNOWAAE	SBB-SNOWJMU
Support Model		IWA-VP(P1.68), IFR/IEA, IWC	IWA, IFA, IFR/IER/IEA	IFA, IFR/IER/IEA
SOC		Pontus-M	Oscar-P	Kant-M2e
Front-end IC		PS196(DP)	SE19HM	SE18S
FPGA		KU060, K200T	KU060, K325T	KU085
Max Screen Resolution		4k(3840x2160)	8k(7680x4320), 4k(3840x2160)	4k(3840x2160)
Connectivity	Input	HDMI(2), DP(1), USB(2)	HDMI(4), DP(1), USB(2)	HDMI(2), DP(1), USB(2)
	Audio Out	Optical (2CH, PCM only)	Optical (2CH, PCM only), eARC	Optical (2CH, PCM only)
	Video Out	QSFP+ x4EA	CWDM x8EA	HDBT x4EA
Output Bit-depth (bit)		10bit (RGB 4:4:4)	10bit (RGB 4:4:4)	10bit (RGB 4:4:4)
Power		78W, DC 19V, 4.1A	180W, DC 24V, 7.5A	78W, DC 19V, 4.1A
OS		TIZEN 6.5	TIZEN 6.0	TIZEN 4.0
Magic Info		MagicINFO S10	MagicINFO S9	MagicINFO S6
Refresh Rate		60Hz	120Hz	60Hz
Sensor		Eco	Eco	Eco
IR		External	External + Internal	External
LAN		RJ45	BT/WiFi, RJ45	RJ45
Standard		EMC Class B	EMC Class B	EMC Class A
Working Temperature (℃)		0˚C~40˚C (32˚F to 104˚F)	0˚C~40˚C (32˚F to 104˚F)	0˚C~40˚C (32˚F to 104˚F)
Operation Humidity		10% ~ 80%	10% ~ 80%	10% ~ 80%

1. Product's Description

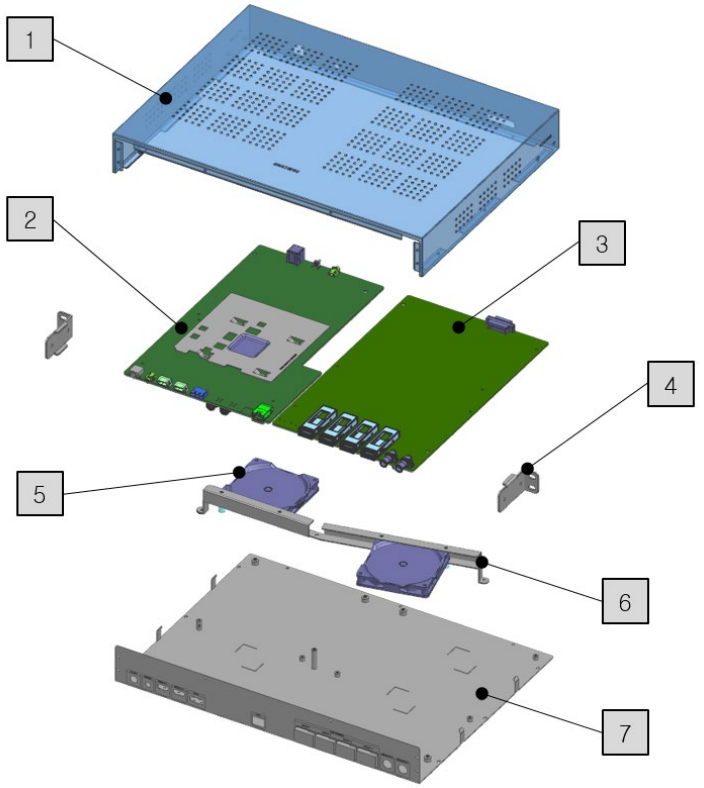
❑ Development Specifications - Mechanical

❑ Product Size Weight

Type	CX4B / CV4B / CS4B
Set Size [WxDxH][mm] (without Foot)	440.0 x 340.0 x 60.5
Weight[kg]	3.95

❑ Mechanical Exploded View

No	Description	Qty
1	ASSY COVER-TOP	1
2	PCB MAIN	1
3	PCB FPGA	1
4	BRACKET-SUPPORT	2
5	FAN	2
6	ASSY BRACKET P-GUIDE	1
7	ASSY BRACKET-PCB	1



1. Product's Description

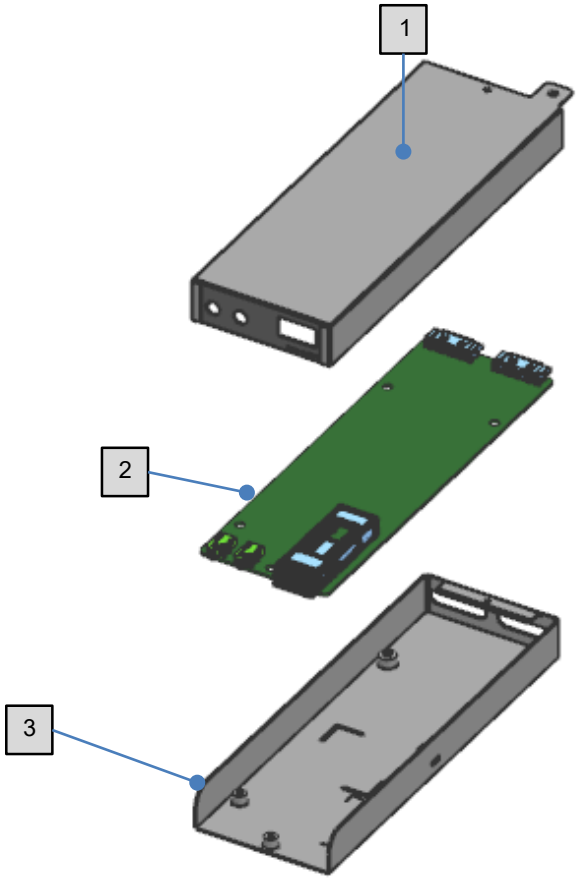
❑Development Specifications - Mechanical - Interface Gender

❑ Product Size Weight

Type	Interface Gender
Set Size [WxDxH][mm] (without Foot)	75.6 x 228.3 x 23.3
Weight[kg]	0.5

❑ Mechanical Exploded View

Type	Description	Qty
1	ASSY COVER-TOP	1
2	PCB SUB-IG	1
3	ASSY BRACKET-PCB	1



2. SBox Install preparations

2. S-Box Install preparations

❑Pre-installation preparations

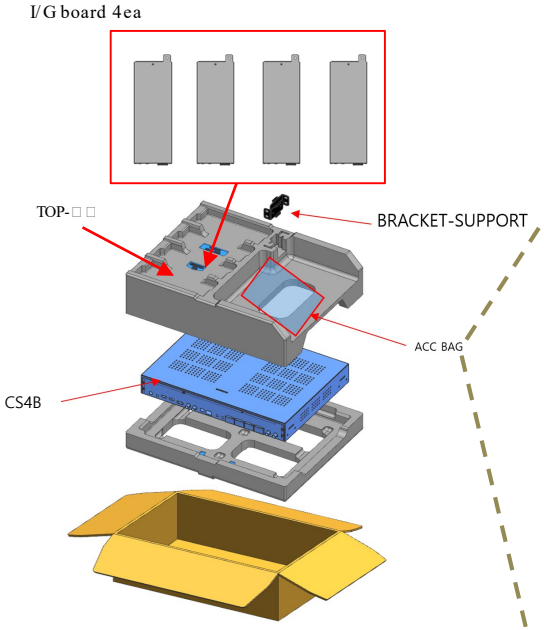
① Remove the tape of the top packing BOX and open the box.

□ Check the following accessory list list while taking out the accessory components loaded in the Top–Cushion.

□ After removing the top-cushion, take out the S-BOX inside the PE BAG and open it.

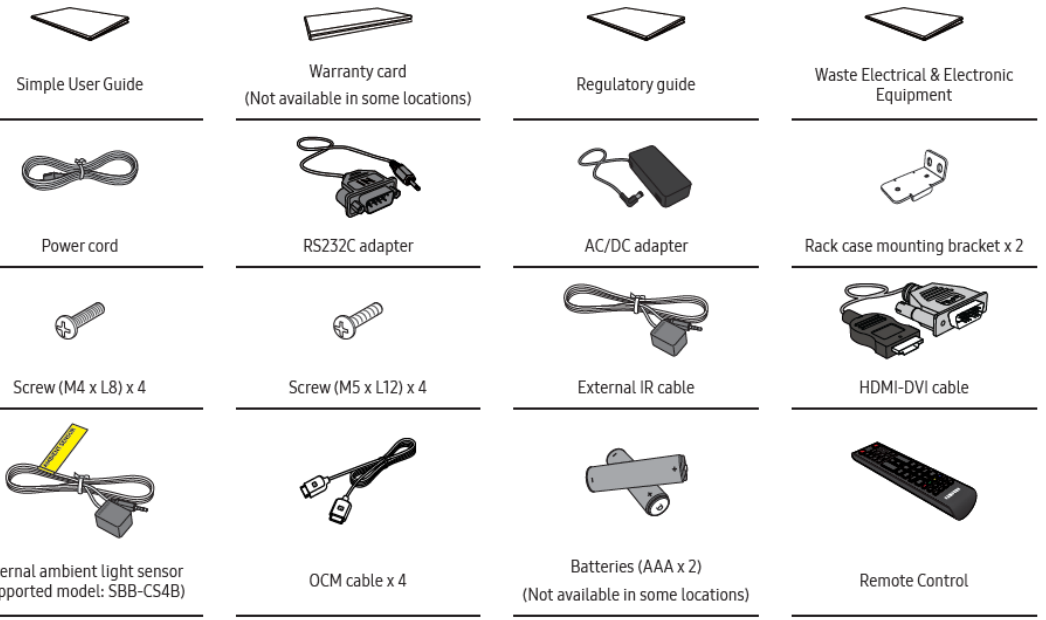
□Power cord by region □

POWER CODE	
US	3903-001117
EU	3903-001118
KOREA	3903-001074
CHINA	3903-001075



Accessory Configuration Part

SBB-CV4B / SBB-CS4B



※ Light Sensor is an accessory that only supports CS4B.

3. S-BOX- Installation and connection

3. S-BOX Installation and Connection

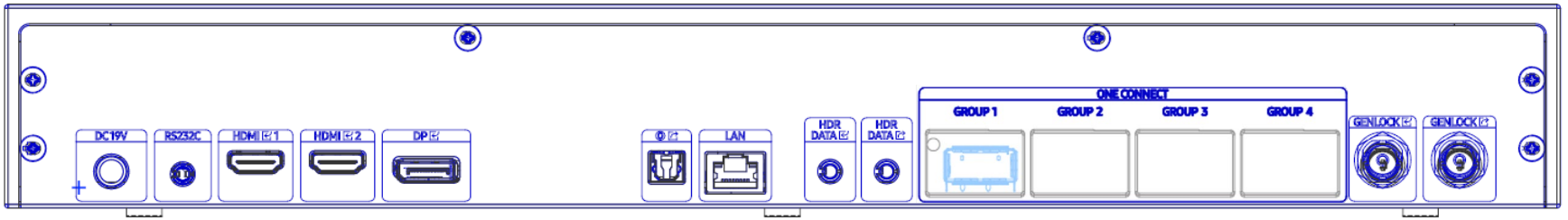
❑ CS4B Front/Back Designation

❑ Front



*The DVI-out is used only for service purposes, so be careful not to use it when connecting the actual cabinet.

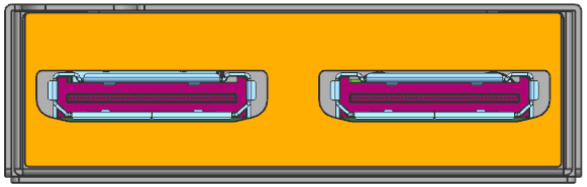
❑ Rear



3. S-BOX Installation and Connection

❑ IG Front/Back Designation

❑ Front



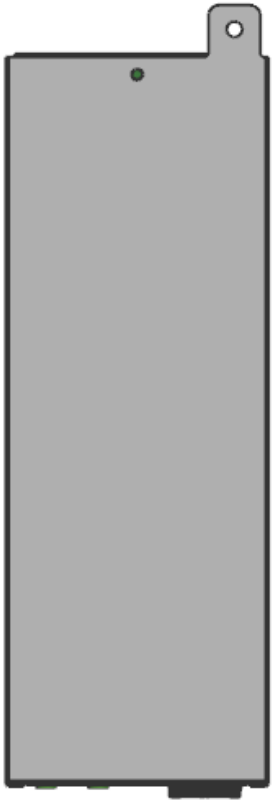
❑ Rear



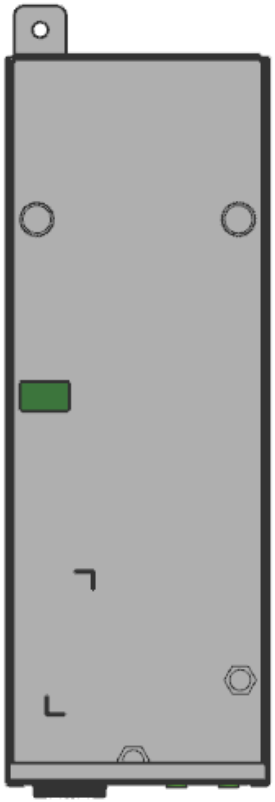
❑ Side



❑ Top



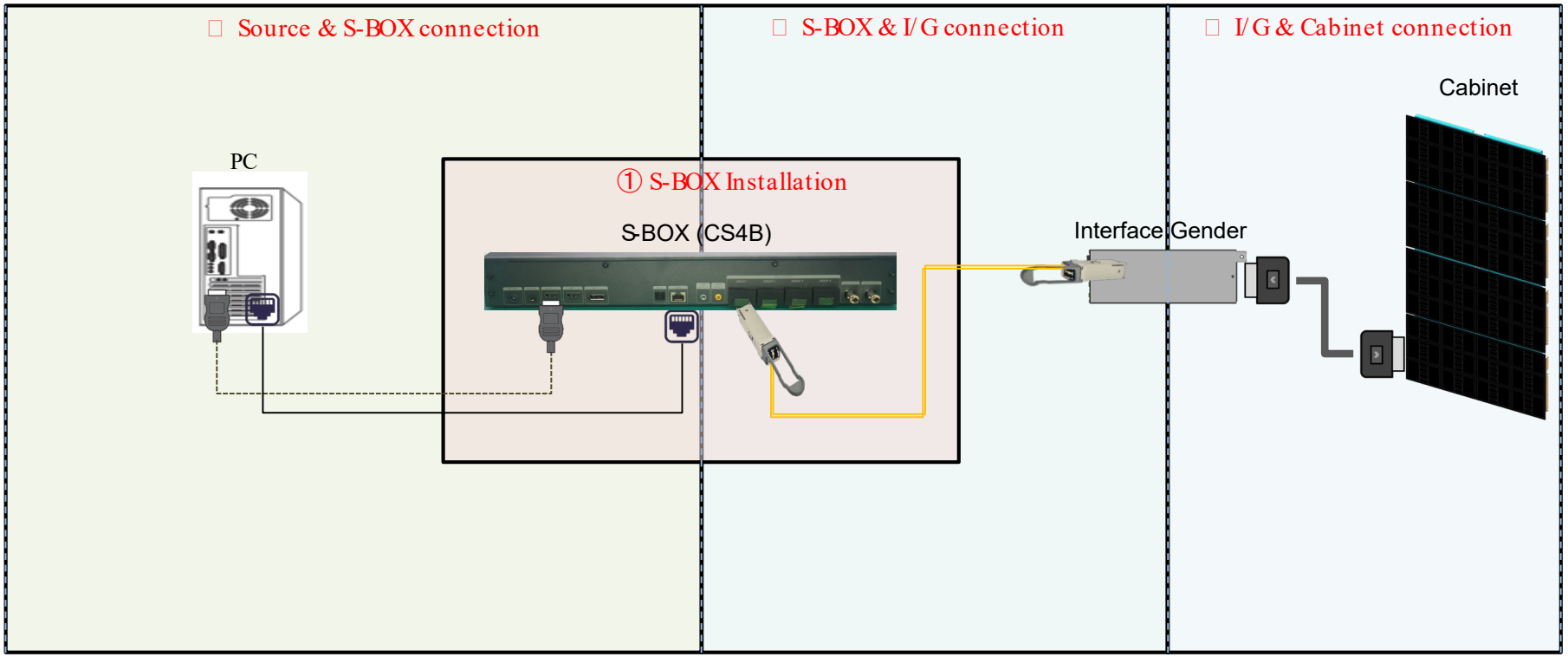
❑ Bottom



3. S-BOXInstallation and Connection

❑ S-BOX Installation Order

- The SBOX installation order proceeds in four steps as follows.



3. S-BOX Installation and Connection

❑ S-BOX Installation Order (①S-BOX Installation)

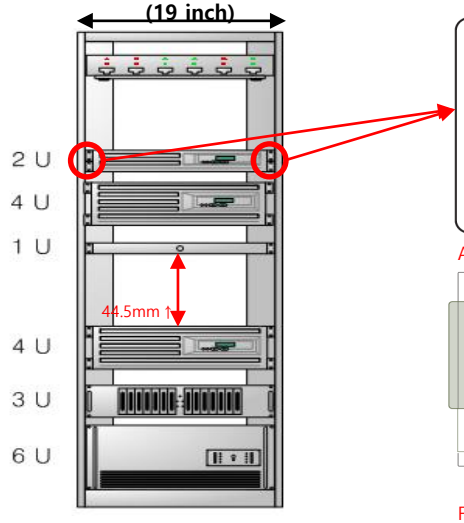
- The SBB-CS4B model is designed to meet the 19inch standard server rack. It is recommended to install on the standard server rack when installing.

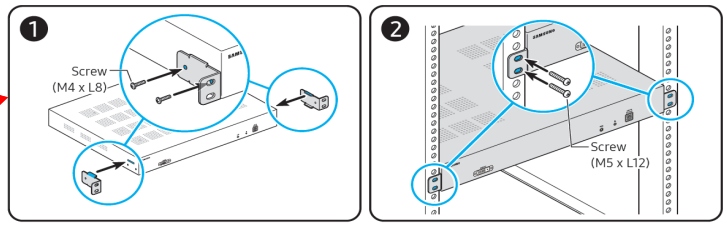
※ When installed on the wall, the product must be more than 10mm away from the wall, up, down, left and right, and in the case of cable port, install it more than 50mm away to connect the cable.

- Keep the residual heat inside the rack below 35°C.
- Install the vents so that they are open, and do not turn sideways or upside down.

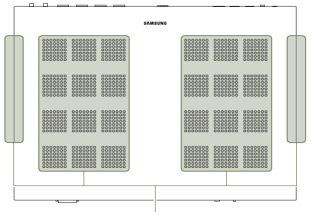
※If you block the vent, the product may overheat, so be careful.

- When installing multiple S-BOXs, install them to be at least 1U (44.45mm) away from the top product.
- Be careful not to let liquids, etc. into the vents of the product..





Assemble using the screw, mounting bracket in the accessory.



For CS4B, the vents are on the TOP side, left/right side. You should be careful not to get stuck.

3. S-BOX Installation and Connection

❑ S-BOX Installation Order (①S-BOX Installation)

- Depending on the screen configuration, you can configure the screen using multiple S-BOXs. (Considering resolution or installation structure of 4K or more)
- When configuring a screen through Multi S-BOX, connect Multi S-BOX using a network router.

※ IP between S-BOX should be set and used differently.



3. S-BOX Installation and Connection

❑ S-BOX Installation Order (①S-BOX Installation)

- SBB-CS4B model is S-BOX that provides LED HDR function.

※ LED HDR: Dynamic Peaking technology implements excellent contrast ratio to express image quality close to real life

- If a screen is configured by connecting multiple S-BOXs, the Multi-Link HDR port of S-BOX must be connected.

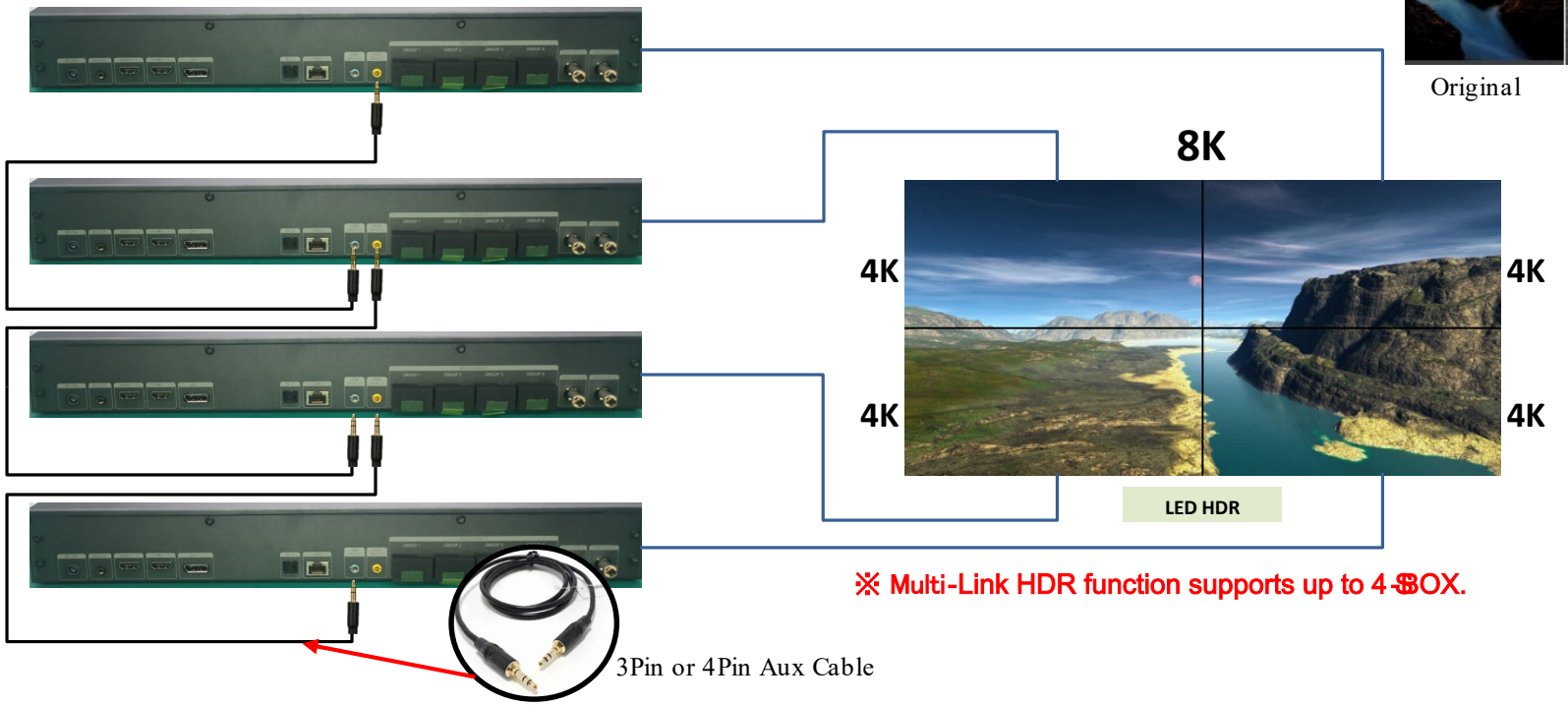
※ When used without connection, the brightness of each connected box may be different due to the gain difference between cabi nets.

- Connect by Daisy Chain method between S-BOX using 3Pin or 4Pin Aux Cable (3.5mm).
- Connect the Aux cable to the Multi-Link HDR port and “ON” the Multi-Link HDR in the S-BOX configuration.

(Refer to page 51)



Original | Max Brigthness | LED HDR



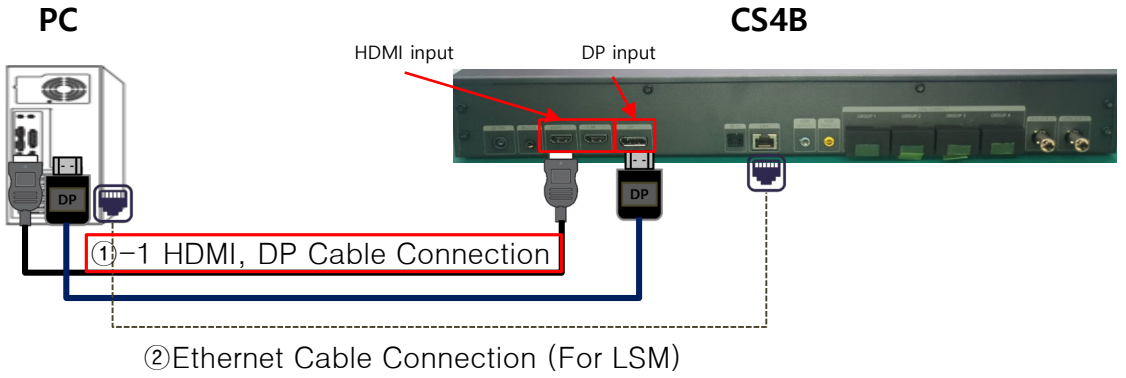
※ Multi-Link HDR function supports up to 4 S-BOX.

3. S-BOX Installation and Connection

❑ S-BOX Installation Order (②Source and S-BOX connection)

- The SBB-CS4B model is an S-BOX that can receive HDMI, DP (Display Port), and SDI (Serial Digital Interface) signals.

① -1) When receiving an input signal to HDMI/DP using a PC, etc., the HDMI cable must be connected using a cable that supports 2.1 specifications / DP cable 1.4 specifications.



3. S-BOX Installation and Connection

❑ S-BOX Installation Order (②Source and S-BOX connection)

② Since the 1920x1080 output is the default output specification, the CS4B model must change the IG resolution setting when connecting to the 1920x2160 resolution support cabinet. (Supportability of 1920x1080 for IFR/IEA and 1920x2160 for IWA-VP/IWC/IVC)

<Support resolution by cabinet>

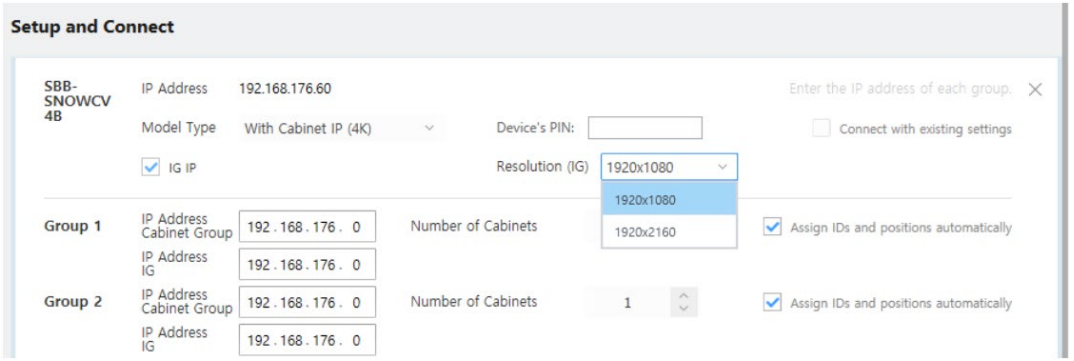
※CS4B does not support IER and IWB

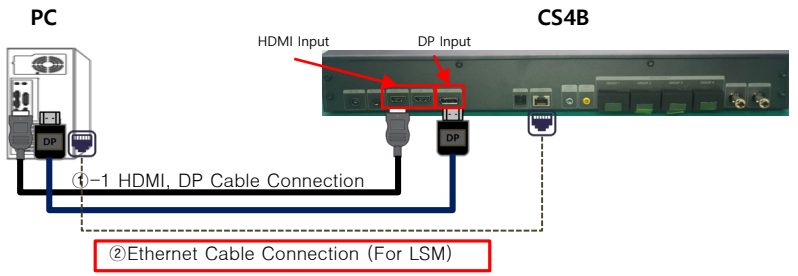
Cabinet		IFR			IEA			IWA			IWC			IVC	
Pitch		P2	P2.5	P4	P2	P2.5	P4	P0.84	P1.26	P1.68	P0.84	P1.26	P1.68	P1.68	P2.1
IG Output	Frame rate	60Hz			60Hz			60Hz			60Hz			60Hz	
	Resolution	1920x1080			1920x1080			1920x**2160**			1920x**2160**			1920x**2160**	
	Compression status	X			X			O			O			O	

1) Connect PC and CS4B using Ethernet Cable (Shield) and connect PC and S-BOX through LSM. (Refer to page 55 for detailed LSM connection guide)

2) In the "Setup and Connect" menu in the LSM, modify the IG Resolution to match the resolution of the connected cabinet.

<Menu in LSM>





3. S-BOX Installation and Connection

❑ S-BOX Installation Order (②Source and S-BOX connection)

※ IG Resolution value is set not to match the supported resolution

If the support resolution of the cabinet connected to the IG Resolution value does not match, the screen may be broken as shown in the picture on the left.

→ This phenomenon is normal for both S-BOX / Cabinet and the screen is broken according to the specifications, so it is not a failure symptom.

→ 19p. If you change the IG resolution through LSM according to the cabinet resolution according to the guide, you can output the normal screen as shown in the picture on the right

In the United States.

Example: S-BOX shipment specification (1920x1080) and IWA (1920x2160 resolution) connection



IG Resolution through LSM
Change to 1920x2160



<When connecting S-BOX shipment specification and IWA>

1920x1080

<When IWA is connected after applying 1920x2160>

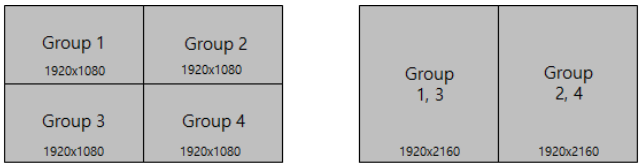
3. S-BOX Installation and Connection

❑ S-BOX Installation Order (②Source and S-BOX connection)



※ Note (1/2)

Please refer to the following picture for the output specification according to the IG Resolution value.



□ When Output Resolution is UHD

1) IG Resolution : 1920x1080



2) IG Resolution : 1920x2160



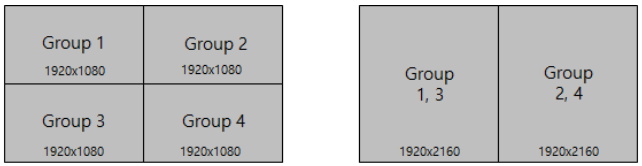
3. S-BOX Installation and Connection

❑ S-BOX Installation Order (②Source and S-BOX connection)



※ Note (2/2)

Please refer to the following picture for the output specification according to the IG Resolution value.

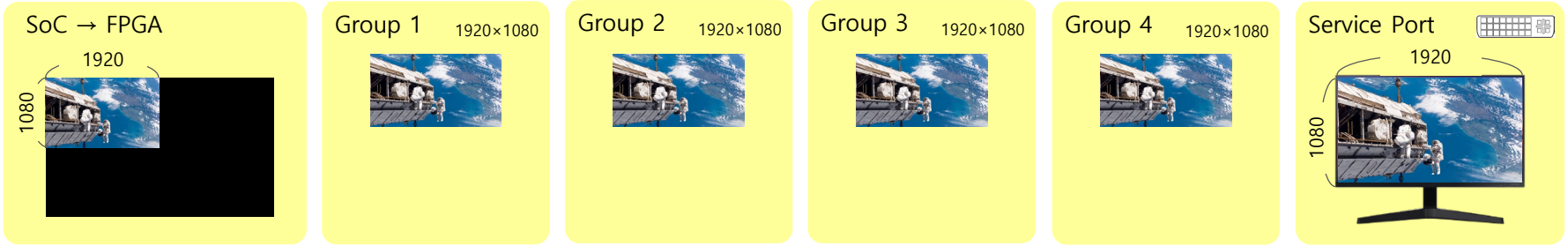


< IG Resolution = 1920×1080 >

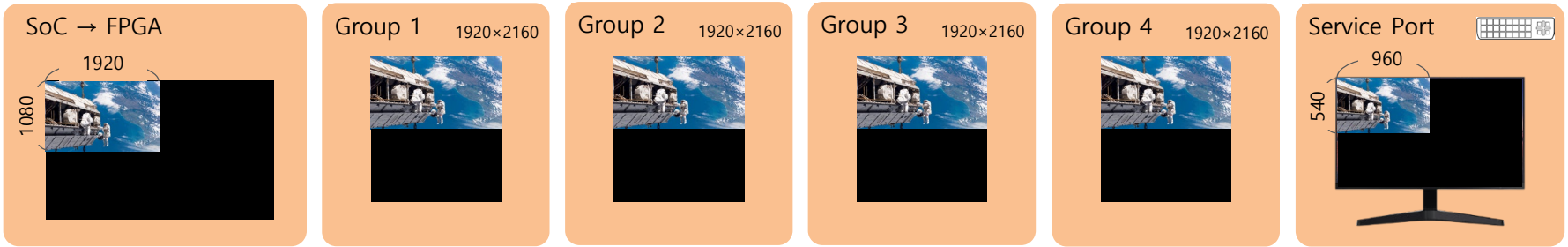
< IG Resolution = 1920×2160 >

□ When Output Resolution is

FHD) IG Resolution : 1920x1080



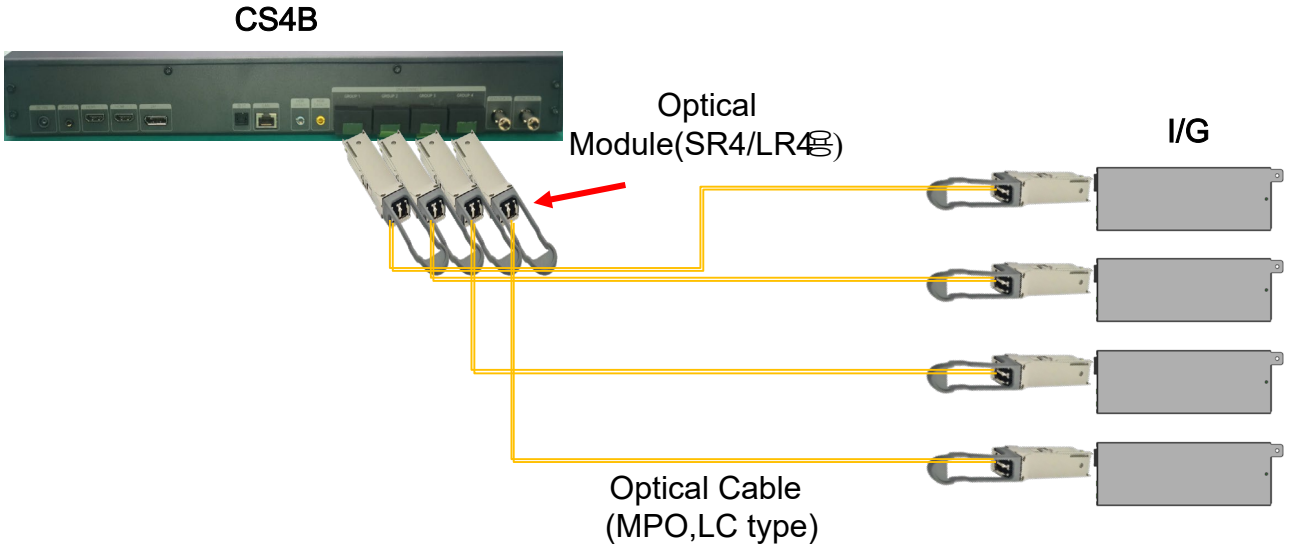
2) IG Resolution : 1920x2160



3. S-BOX Installation and Connection

❑ S-BOX Installation Order(③S-BOX and I/G Connection)

- Connect S-BOX and I/G using optical module (for SR4/LR4) and optical cable (MPO,LC type).



3. S-BOX Installation and Connection

❑ S-BOX Installation Order(③S-BOX and I/G Connection)

① Optical module connection

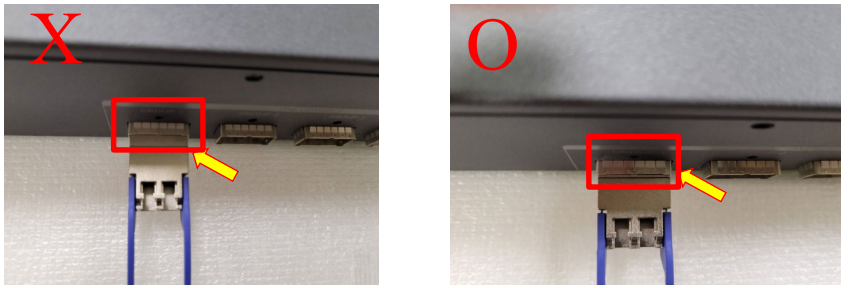
- CS4B is a product that can be connected using optic transceiver (optic module) that supports 40G QSFP+ specification.
- Connect the optical module to the S-BOX and I/G connectors using the optical module (for SR4 or LR4).

Short-range transmission (100m) Long-range transmission (10km)

LR4 SR4



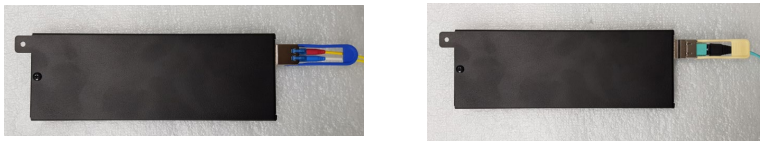
※ When connecting the optical module, push it until it clicks.



<Photo of optical module and SBOX connection>

※ CS4B is compatible with both QSFP + SR4 / LR4 optical modules.

LR4 SR4



<Photo of optical module and I/G connection>



<SR4 Optical Module> <LR4 Optical Module >

※ During S-BOX operation, do not remove or connect the optical module. It can cause failure.

3. S-BOX Installation and Connection

❑ S-BOX Installation Order(③S-BOX and I/G Connection)

① Optical module connection – Precautions

- The optical module specification compatible with CS4B is QSFP+40GBASE-LR4/SR4, and it does not work when connecting optical modules of other specifications.
- The following list is an optical module compatible with the CS4B model, so it is recommended to use the optical module by referring to the following list..

MAKER	PART NUMBER	CATEGORY	WAVELENGTH	REACH	INTERFACE	FIBER TYPE
FIBERATE	QSMM854T01D	SR4	850nm	100m	MPO	MM
FIBERATE	QSSL274T10D	LR4	1310nm	10km	LC	SM
FS	QSFPSR440G	SR4	850nm	100m	MPO	MM
FS	QSFPLR440G	LR4	1310nm	10km	LC	SM
RUIGU	40GBASELR4	LR4	1310nm	10km	LC	SM
RUIGU	40GBASESR4	SR4	850nm	100m	MPO	MM
EXA	EXA-40G-QC01	SR4	850nm	100m	MPO	MM
Finisar	QSFPSR440G	SR4	850nm	100m	MPO	MM
MENTECH	QSFPSR440G	SR4	850nm	100m	MPO	MM
MENTECH	QSFPLR440G	LR4	1310nm	10km	LC	SM
QSFPTEK	QT-QSFP+-LR4	LR4	1310nm	10km	LC	SM

3. S-BOX Installation and Connection

❑ S-BOX Installation Order(③S-BOX and I/G Connection)

① Optical Module Connection – Notes

- For LR4 / SR4 specifications, refer to the following IEEE 802.3ba specifications.

Parameter		Min.	Typ.	Max.	Unit	Reference
Supply voltage		3.135	3.3	3.465	V	3.3V ± 5%
Power consumption		-	-	3.6	W	1.15A
LR4	Tx average launch power, each lane	-7	-	2.3	dBm	802.3ba 87.7.1
	Tx optical modulation amplitude, each lane	-4	-	3.5	dBm	802.3ba 87.7.1
	Rx average receive power, each lane	-13.7	-	2.3	dBm	802.3ba 87.7.2
	Rx receive sensitivity(OMA), each lane	-	-	-11.5	dBm	802.3ba 87.7.2
SR4	Tx average launch power, each lane	-7.6	-	2.4	dBm	802.3ba 86.7.1
	Tx optical modulation amplitude, each lane	-5.6	-	3	dBm	802.3ba 86.7.1
	Rx average receive power, each lane	-9.5	-	2.4	dBm	802.3ba 86.7.2
	Rx receive sensitivity(OMA), each lane	-	-	-5.4	dBm	802.3ba 86.7.2

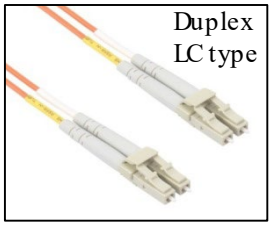
※IEEE 802.3ba specification

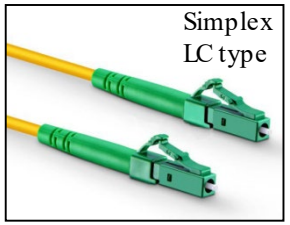
3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order(③S-BOXand I/G Connection)

② Optical Cable connection

- When installing the optical module with LR4 specification, connect it using LC Type SMF specification optical cable.
  (Single Mode Fiber)
- When installing the optical module with SR4 specification, connect using MPO Type MMF specification optical cable.
  (Multi Mode Fiber)
- When connecting using LC type, release the locking structure when connecting using Duplex type cable and connect one by one like Simplex.



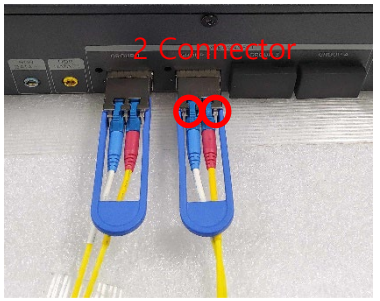




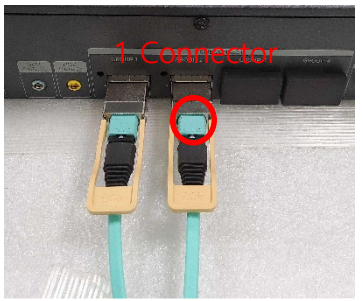
- Be careful not to contaminate the ends before insertion and connect them with a soft cloth for cleanliness.



<Prevention of contamination at the end of cable>



Connect two connectors when connecting LC type



Connect one connector when connecting MPO type

3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order(③S-BOXand I/G Connection)

② Optical cable connection (connection example)

When connecting LR4



When connectingSR4



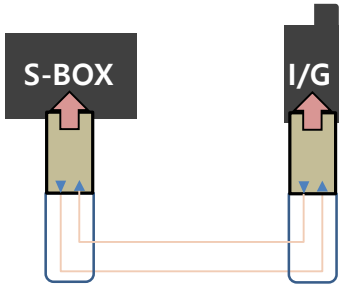
When connecting LR4



When connecting SR4



<S-BOX연결 사진>



When connecting to the LC type cable, the transmission/reception unit of the S-BOX and IG optical modules must be checked and connected.

TRANSMISSION UNIT OF S-BOX OPTICAL MODULE -> RECEIVING UNIT OF IG OPTICAL MODULE
RECEIVING UNIT OF S-BOX OPTICAL MODULE -> TRANSMITTING UNIT OF IG OPTICAL MODULE

※ If the screen is not turned on or communication is not smooth, change the left and right of the LC Type connector inserted into the optical module.



<I/G connection photo>

3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order(③S-BOXand I/G Connection)

② Optical cable connection - precautions

- When connecting signals using optical cables, handling defects may occur, so it should be installed in consideration of the following optical cable handling specifications.

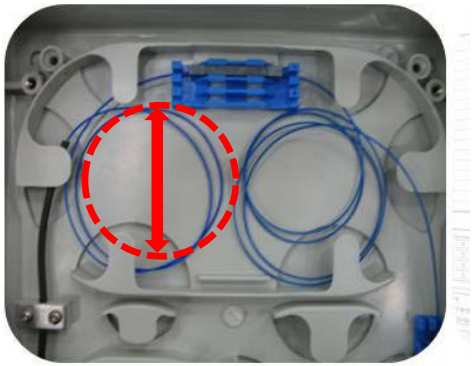
- Avoid contact devices with sharp or rough ends.
- Do not wipe the optical cable with coarse material or dissolving agent.

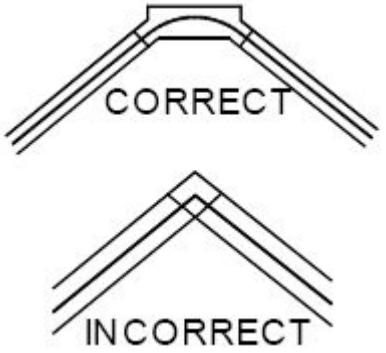
Tensile force characteristics vary depending on the specifications of the cable used.
Please refer to the cable specifications you use.

- Unconnected fiber optic cable connectors must use a protective cap and inspect and clean the connectors before connection.
- Install the cable so that the minimum force is applied to the optical fiber in consideration of the pulling force.
- Optical cables should not be stepped on with feet or pressed by heavy water. There is a concern that the cable may be damaged by the weight.
- Cable ties, etc. should not be used when installing. Loss or fiber optic disconnection due to compression and flexion may occur.
- Bending the optical cable above the allowable bend radius causes optical loss, so it must be installed below the allowable bend radius. (Refer to the figure)

The allowable bend radius varies depending on the specifications of the cable used.
Please refer to the cable specification.

- Be careful not to bend or twist at right angles. (Refer to the figure)







3. S-BOXInstallation and Connection

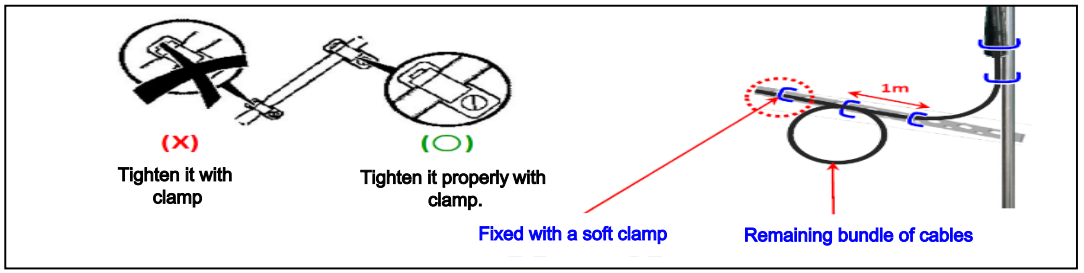
❑ S-BOXInstallation Order(③S-BOXand I/G Connection)

② Optical cable connection - precautions

- When connecting signals using optical cables, handling defects may occur, so it should be installed in consideration of the following optical cable handling specifications.

Compression characteristics vary depending on the specifications of the cable used.
Please refer to the cable specifications you use.

- Fix it to Tray, etc. and gently fix it so that it is not pressed directly on the optical cable when used.
- The clamp spacing used for fixing is recommended to be less than 1m.
- Do not impact the optical cable. The impact load of a typical optical cable is 7J or less, but the impact characteristics should be referred to each cable specification.
- Be careful not to cause cable torsion. If the cable twists, it can cause damage to the fiber optics.





- When using FDF (light distributor), there may be deterioration of the signal due to connection loss and reflection loss, so the product does not guarantee it.



FDF, optical distributor

3. S-BOX Installation and Connection

❑ S-BOX Installation Order(③S-BOX and I/G Connection)

② Optical cable connection - precautions

- The CS4B model is a model that supports 40G BASE-SR4/LR4.
- When transmitting an optical signal using the SR4 optical module, it must be connected using the Multi-Mode OM3 Cable (MPO Type). (MAX installation length: 100m)
- When transmitting an optical signal using the LR4 optical module, it must be connected using a single mode cable (LC type). (MAX installation length: 10km)
- Please refer to the following table for cable related specifications.

Interface Type	Fiber Optic Patch Cable Type
1000BASE-SX	Multimode OM1, OM2 Cable
1000BASE-LX	Single-mode Cable
1000BASE-EX	Single-mode Cable
1000BASE-ZX	Single-mode Cable
10GBASE-SR	Multimode OM3, OM4 Cable
10GBASE-LR	Single-mode Cable
10GBASE-ER	Single-mode Cable
10GBASE-ZR	Single-mode Cable
40GBASE-SR4	Multimode OM3, OM4 Cable
40GBASE-LR4	Single-mode Cable
40GBASE-ER4	Single-mode Cable
100GBASE-SR10	Multimode OM3, OM4 Cable
100GBASE-SR4	Multimode OM3, OM4 Cable
100GBASE-LR4	Single-mode Cable
100GBASE-ER4	Single-mode Cable

<Cable type that can be connected by interface>

Mode	Category	100M (FX)	1G (SX)	10G (SR)	40G SR4	100G (SR10)
Single Mode	OS1	10~120Km				
		Less than 1dB/ km				
	OS2	10~120km				
		Less than0.4dB / km				
Multi Mode	OM1	2Km	275m	33m	X	X
	OM2		550m	82m	X	X
	OM3			300m	100m	100m
	OM4			400m	150m	150m
	OM5					

<Transmission distance by cable type>



Single Mode Fiber (LC Type)

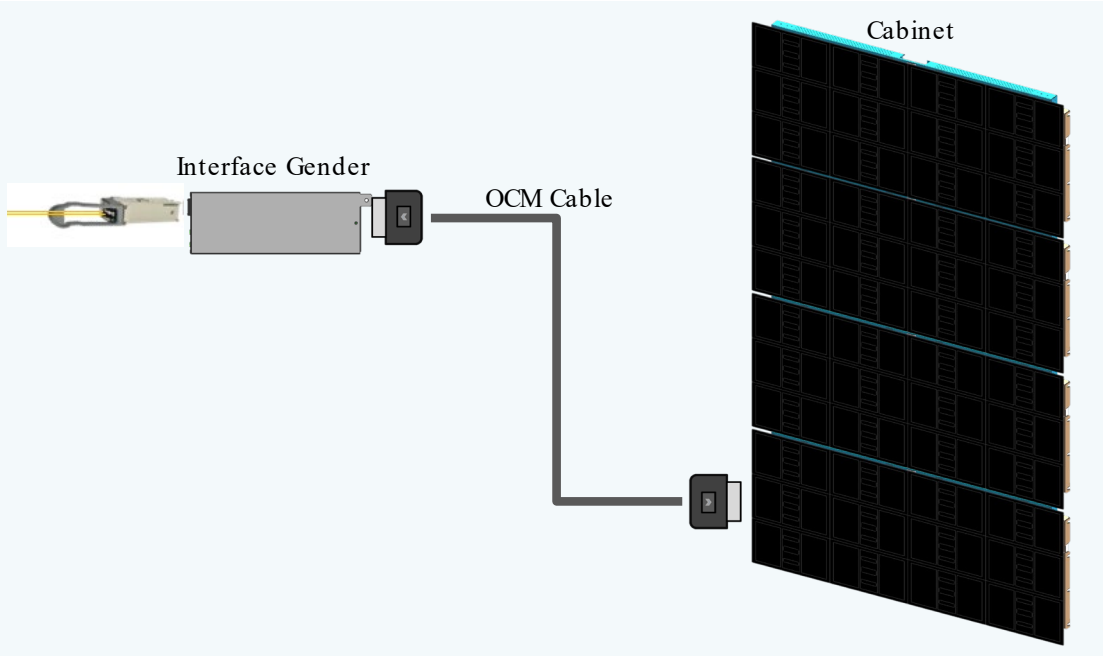


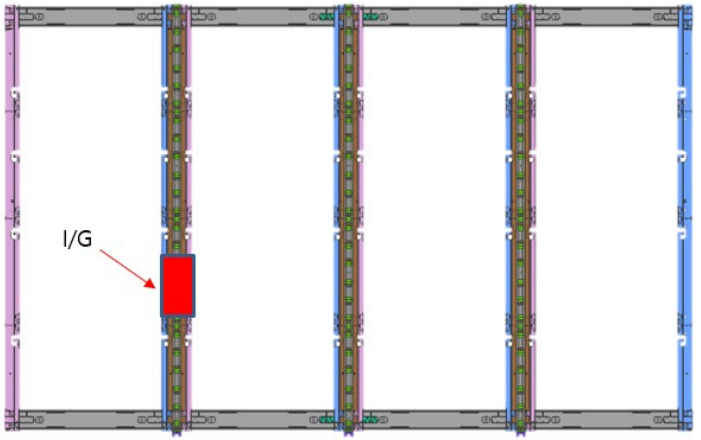
Multi Mode Fiber (MPO Type)

3. S-BOX Installation and Connection

❑ S-BOX Installation Order (④ I/G and Cabinet Connection)

- The CS4B model is an S-BOX compatible with IWA-VP (P1.68), IFREA, IWC Cabinet.
- Connect the I/G and cabinet using the OCM cable in the accessory box.
- I/G is a specification that is installed in the cabinet frame together. Connect the optical cable first and then connect it to the frame so that it can be installed.
- ※ Please refer to the LED Cabinet manual for detailed installation specifications.





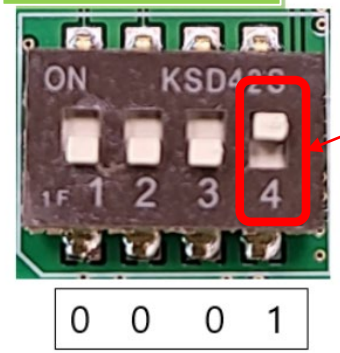
For IWA, I/G is installed in Cabinet Frame.

3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order (④I/G and Cabinet Connection)

- I / G factory shipment Dip Switch specification is 0000. (Refer to the following "shipping mode" photo)
- When installing with LED cabinet,connect standby pin 4 should be turned on and installed with 0001 specificationto enter Always ON modeat all times. (Refer to the following “Always ON mode“ photo)

Always On mode



Connection Standby Pin

In the always-on mode, it enters the always-on state regardless of whether there is an S-BOX signal or not.

※ Consumer installation recommended specifications

Shipping mode

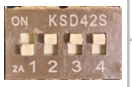


When in shipment mode, enter stand by mode according to the signal of S-BOX.

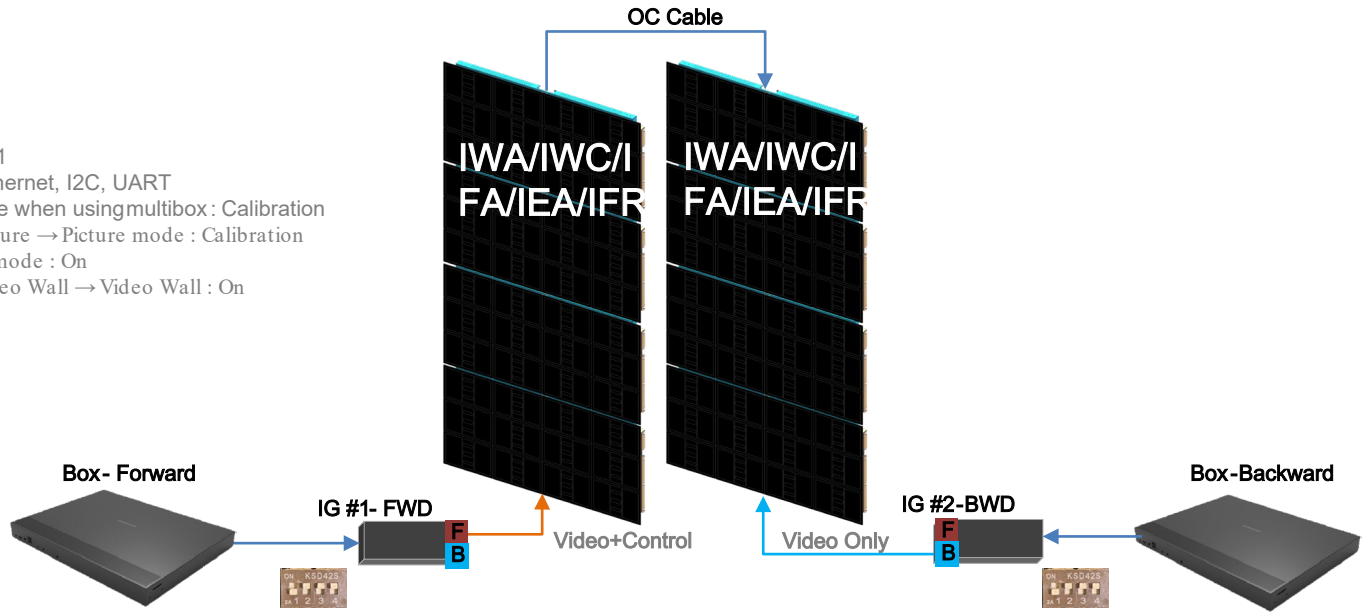
※If no signal is received from the SBOX for 5 minutes after the cabinet power is applied during the shipment mode, it enters the standby mode.

3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order (④I/G and Cabinet Connection)

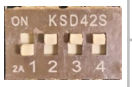
Screen structure	IG Setting	Dip SW Setting	IG Port	Video (Vx1)	Ethernet	STB_EN	Redundancy SW (UART)	PILLAR_DET	Usage
Cable	FWD/BWD IG same		FWD	O	O	O	O	N/C	Cable Installation FWD/BWD Configuration
			BWD	O	X	X	X	X	

- Video : Vby1
- Control : Ethernet, I2C, UART
- Picture mode when using multibox : Calibration
  Menu → Picture → Picture mode : Calibration
- Video Wall mode : On
  Home → Video Wall → Video Wall : On



3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order (④I/G and Cabinet Connection)

Screen structure	IG Setting	Dip SW Setting	IG Port	Video (Vx1)	Ethernet	STB_EN	Redundancy SW (UART)	PILLAR_DET	Usage
Cable	FWD/BWD IG same		FWD	O	O	O	O	N/C	Cable Installation FWD/BWD Configuration
			BWD	O	X	X	X	X	

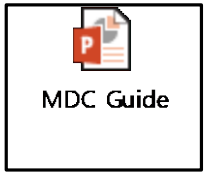
3. S-BOXInstallation and Connection

❑ S-BOXInstallation Order (④I/G and Cabinet Connection)

- IG setting can be adjusted through MDC command.

No.4	MDC Command (Always on mode)	Operating Specification
	Not supported	Always On Mode set
	0 : AA d0 02 02 54 00	Shipment mode (Standby mode entry specification)
	1 : AA d0 02 02 54 01	Always On Mode set

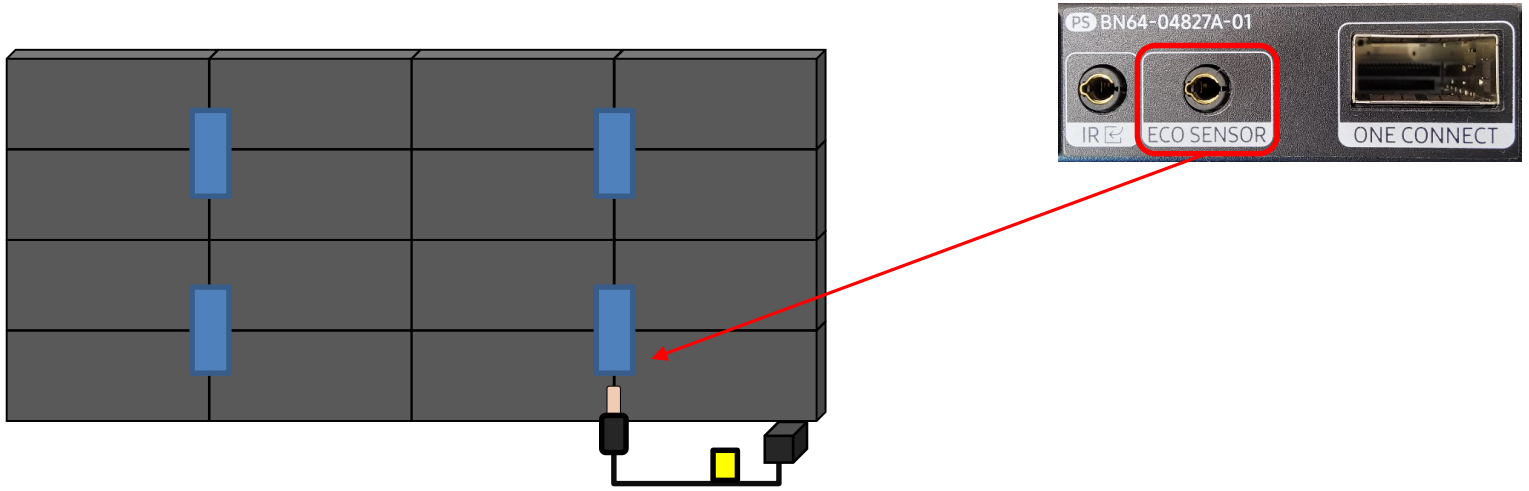
■ If pin No.4 is off, Always on/off control is possible through MDC command
※If pin 4 is on, it always works in on mode. (MDC is not supported)
※Refer MDC Guide



3. S-BOX Installation and Connection

❑ S-BOX Installation Order (④ I/G and Cabinet Connection) – Eco Sensor Installation

- The CS4B model can adjust the brightness of the cabinet according to the brightness change of the external lighting using the illuminance sensor (BN96-39535B). (※ Illuminance sensor is provided as accessory)
- When using the illuminance sensor, connect the interface gender to the “ECO SENSOR” connector.
- Only one illuminance sensor is provided per S-BOX, and it can be used by connecting to 1” IGs.
- It is not available in Multi S-BOX environment, and to use the function of illuminance sensor, you must reboot S-BOX after connecting to IG.
- The illuminance sensor function can be turned on/off through LSM. (Refer to 68 Page)

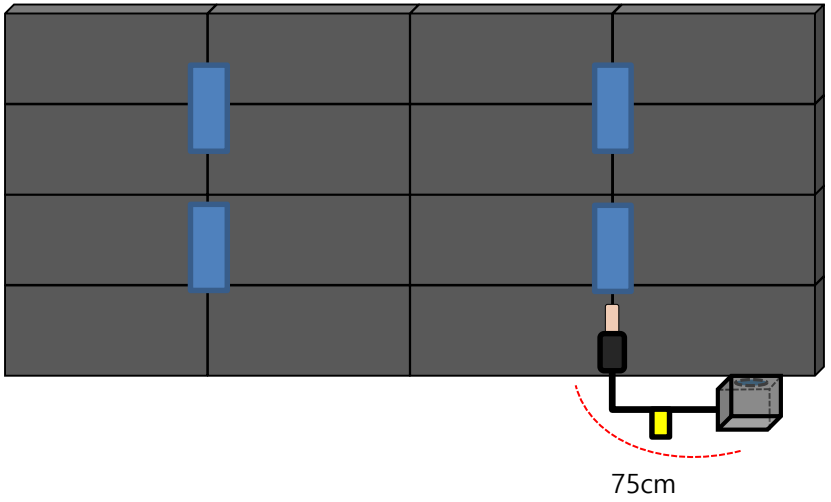


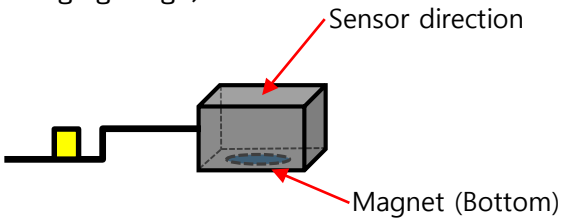
3. S-BOX Installation and Connection

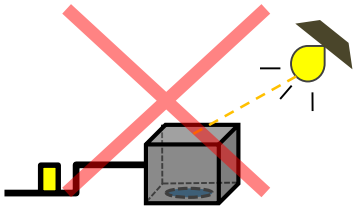
❑ S-BOX Installation Order (④ I/G and Cabinet Connection) – Eco Sensor Installation precautions

- The illuminance sensor is a structure in which a magnet is attached to the sensor part and Bottom to the TOP.
- When installing the illuminance sensor, be careful not to receive direct sunlight from external light.
- The cable length of the illuminance sensor is about 75cm, so it is recommended to connect to the outer IG.
- When installing an illuminance sensor, it is recommended to install it at the bottom.

(※ The installation location may vary depending on the installation environment and surrounding lighting.)







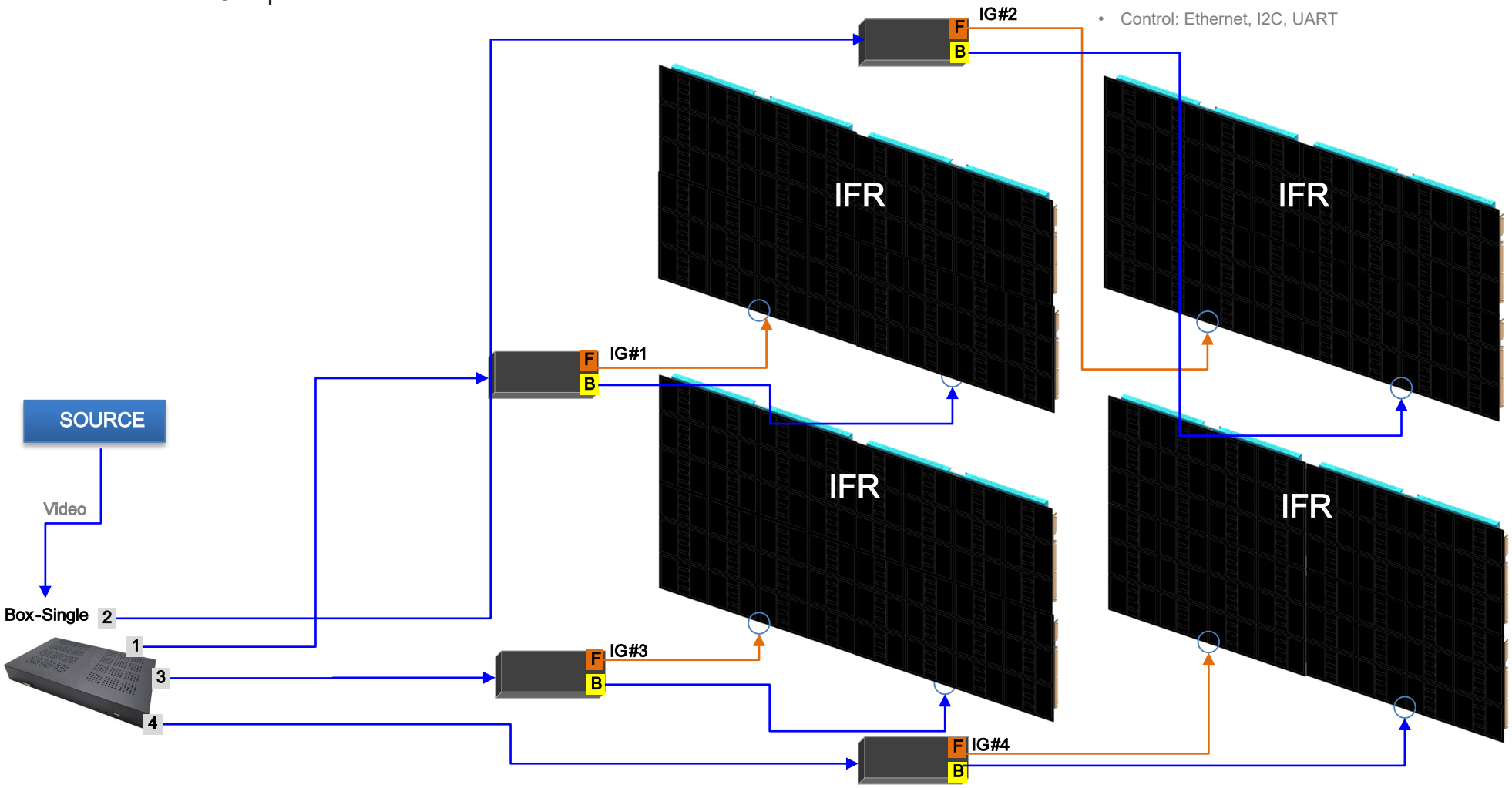
3. S-BOXInstallation and Connection

❑ S-BOX Installation Order④I/G and Cabinet Connection

※ The following connection diagram is Video Redundancy specification, Backward direction does not support data path for control operation.

- IFR Screen Composition

- Video: VBY1
- Control: Ethernet, I2C, UART



3. S-BOX Installation and Connection

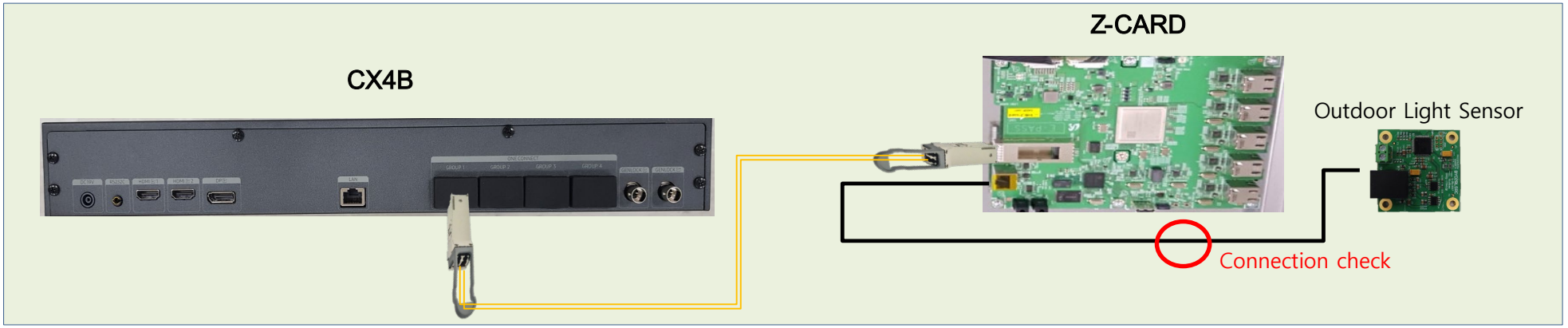
❑ S-BOX Installation Order (④I/G and Cabinet Connection) – CX4B/CS4B

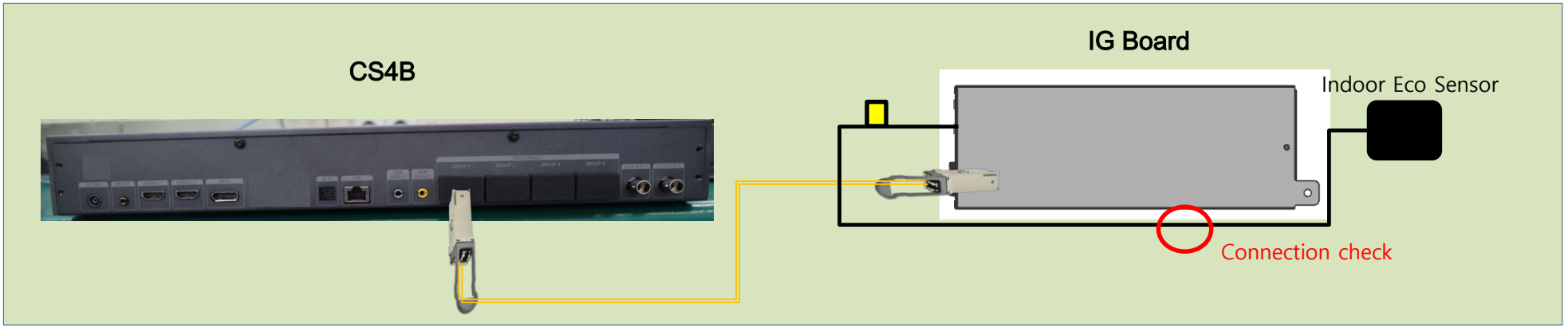
- If the screen brightness is controlled using the Light/Eco Sensor, the screen brightness control operation may not be performed in case of communication failure with the Sensor.

  * Dimming-Brightness Optimization Control through LSM

  * When a communication failure occurs with the sensor, the brightness is fixed to the default value.

- If brightness control through the sensor does not work, reboot the S-BOX (CX4B/CS4B) after reconfirming the cable and sensor installation specifications connected to the sensor. Normal operation is possible only when it is done.





3. S-BOX Installation and Connection

❑ Note) DVI output function for service

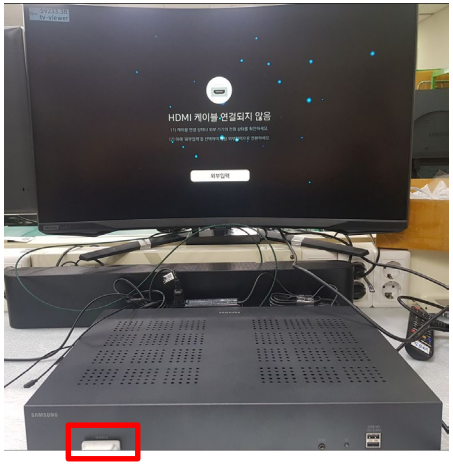
- The DVI-OUT on the front of the product is a port used for service purposes.
- The resolution of the port is FHD (1920*1080 @ 60Hz) and screen blinking and breaking may occur when entering the source of UHD resolution into S-BOX,

This is a phenomenon in which the port output is fixed down scaling without a specific scaling algorithm and has nothing to do with the actual LED cabinet screen output.

- Connect the monitor to use the SERIVCE port.



Actual output image

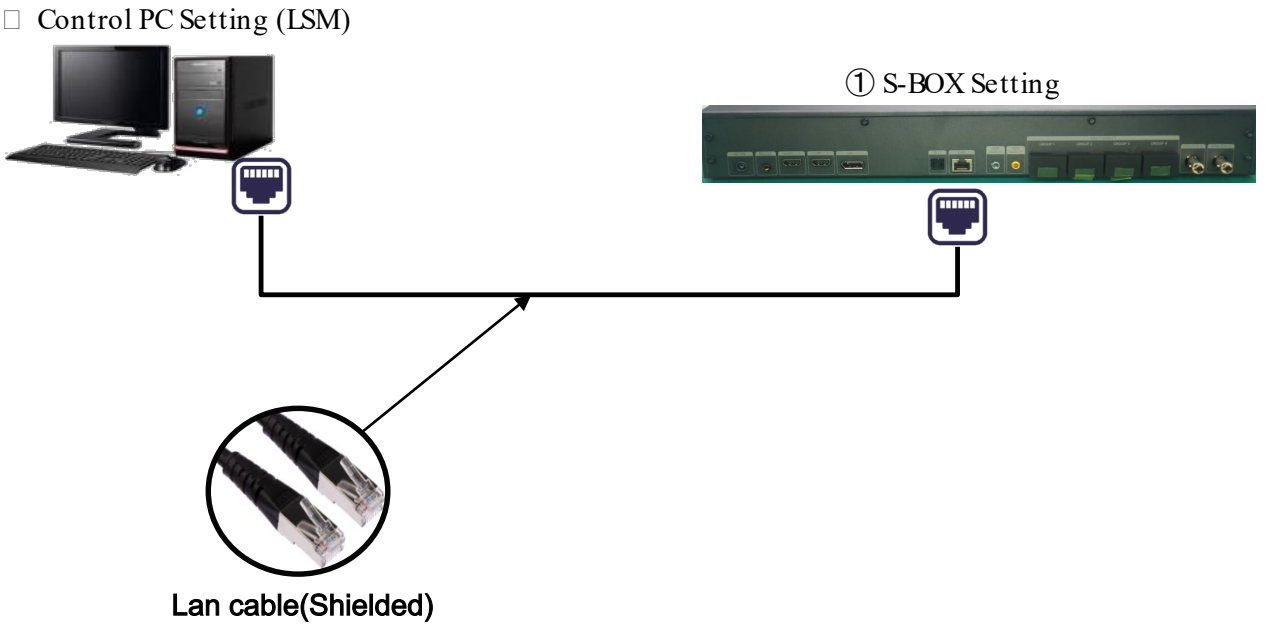


4. Setting and guide

4. Setting andguide

❑ Setting order

- Set 1 S-BOX setting 2 Control PC setting (LSM) in order to use S-BOX.
- Connect using Lan Cable (Shield) between S-BOX ↔ Control PCs.



4. Setting andguide

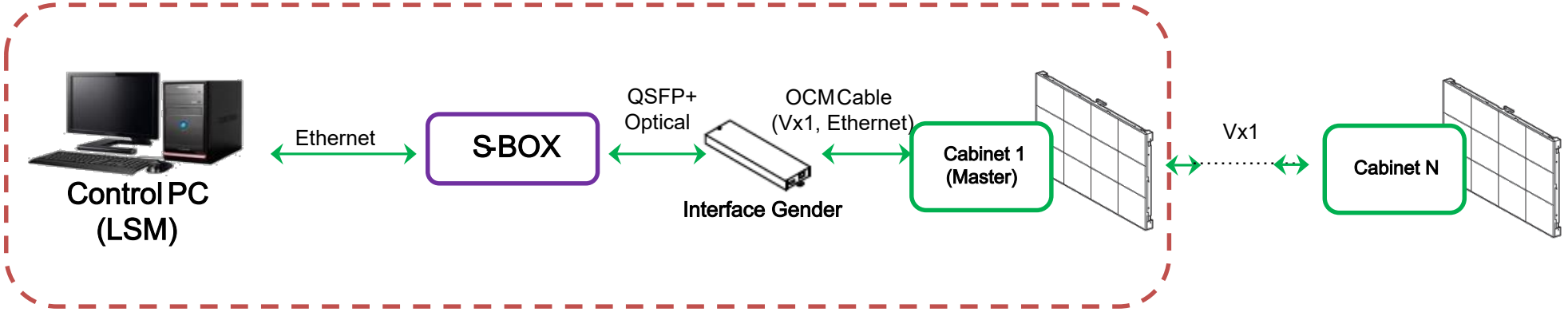
❑ Precautions Before Setting

- S-BOX shipment quality setting is optimized for cabinet specifications and factory shipment, and control BOX ↔ BG must be set for optimal image quality

↔Cabinet should be connected to Optical / OCM cable and installed so that it can be set through LSM.

※The color may be unnatural when the cabinet is lit after LSM is not connected between S-BOX and LED display.

- The image quality setting criteria are set based on the Master Cabinet model connected to Group1 of S-BOX.

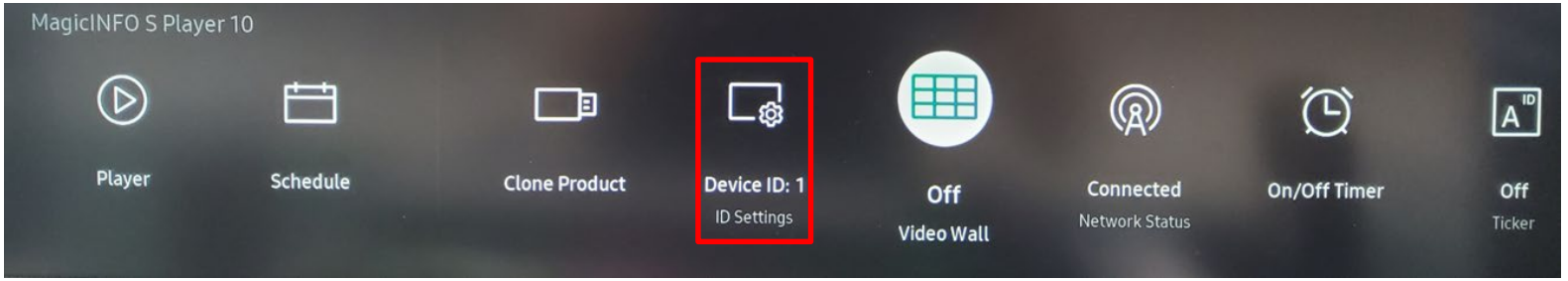


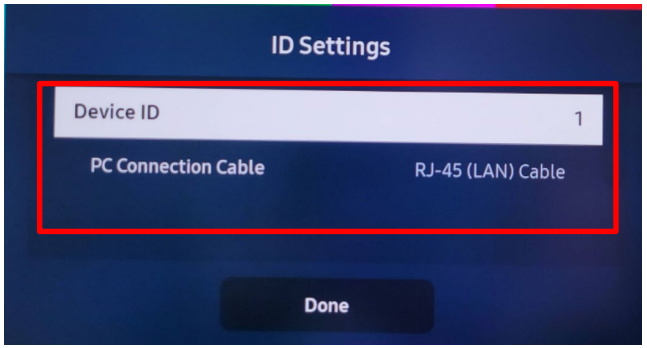
4. Setting andguide

❑ ① S-BOXSettings– Network Settings

- Enter the Home key → Device ID using the remote controller.
- Change the device ID to “1” and the PC Connection Cable to “RJ45 (LAN) Cable”.

For the SBB model, the device ID is always 1.





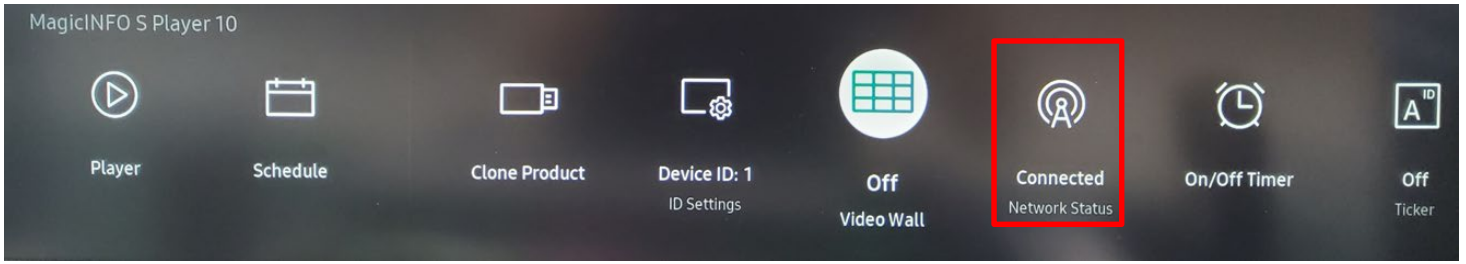
4. Setting andguide

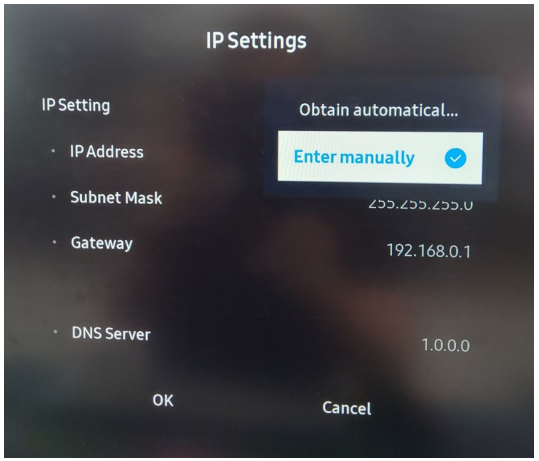
❑ ① S-BOXSettings– Network Settings

- Use Remote Controller to enter Home key → Network Status.You can set up automatic IP or manual IP through IP Setting.

-Set the IP according to the control PC and network environment you use.

※ Recommend manual IP settings.





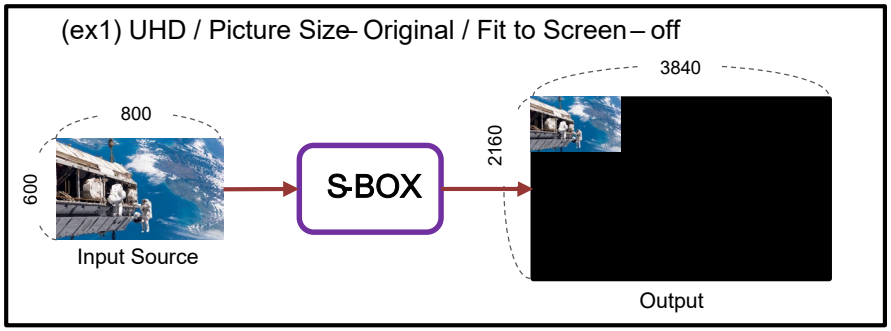
4. Setting andguide

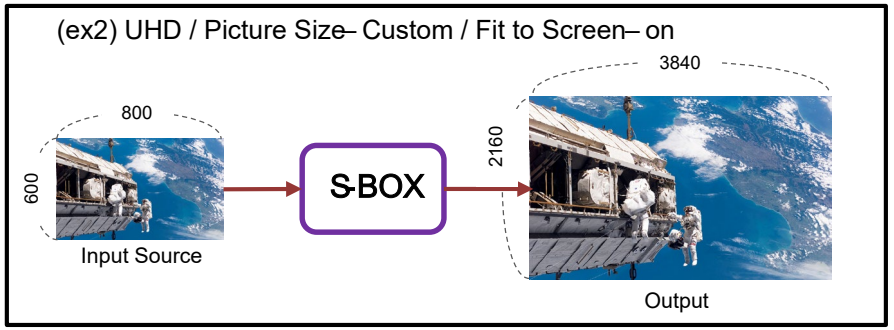
❑ ① S-BOX Settings– Picture Setting

- You can set Menu → Picture specifications using Remote Controller.
- You can set the screen size to display as Cabinet through “LED Picture Size Settings”.

<LED Picture Size Settings>

Output Resolution	UHD	Select UHD/FHD to match the resolution of the source entered by the Control PC.
	FHD	
LED Picture Size	Original	Output resolution is set to match the source ratio of the input source.
	Custom	Output resolution is set by setting the desired width/height (ratio).
Fit to Screen	Off	Regardless of the LED cabinet size, it is displayed according to the original ratio of the input source.
	On	Edit and print the image ratio to fit the LED size.





4. Setting and guide

❑ ① S-BOX Settings – Picture Setting (Exceptions)

- Picture Size of S-BOX does not provide overscan/ crop function by default.
- If the LED Picture Size is Original or Fit to Screen is Off, the original ratio is maintained. If it is Fit to Screen On, it is adjusted to the LED Picture Size.

Displays the screen with Up scaling/Down scaling.

- If the input resolution is 4096x2160, the output resolution is set by cropping the following exceptions.

< Exceptions>

Case 1

Output Resolution : UHD (3840 x 2160)

LED Picture size : Original

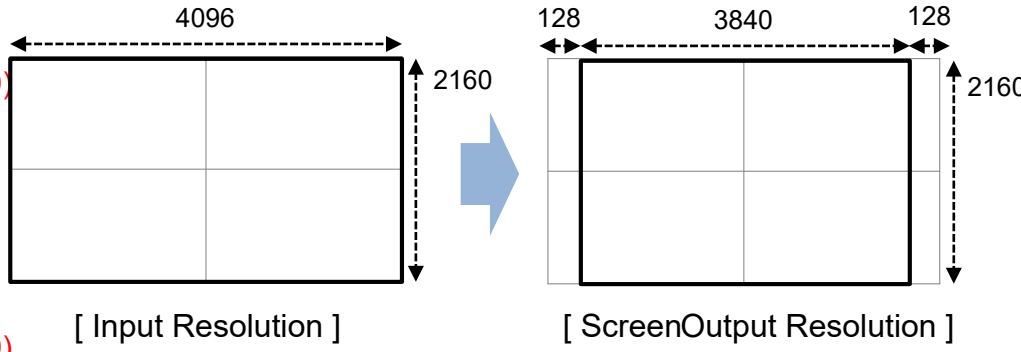
→ Crop the horizontal size and output it. (Crop size : (4096-256) x 2160)

Case 2

LED Picture Size : Custom (3840 x 2160)

Fit to Screen : Off

→ Crop the horizontal size and output it. (Crop size : (4096-256) x 2160)



[Input Resolution] [Screen Output Resolution]

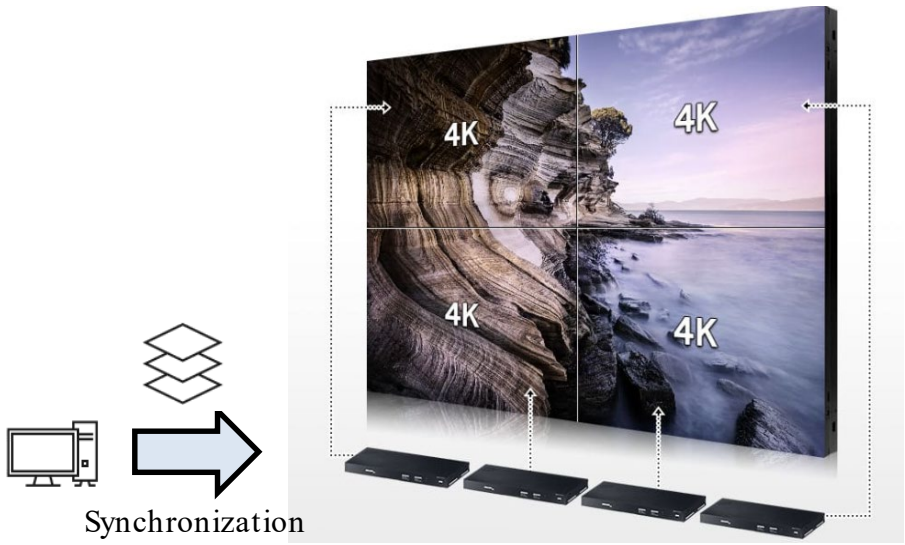
4. Setting andguide

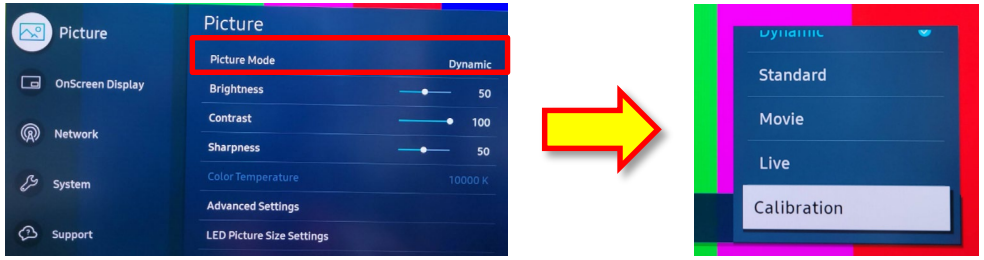
❑ ① S-BOX Settings– Grouping setting

- You can configure a single screen using multiple S-BOXs using the grouping function of S-BOX.
- To use the grouping function, change the image quality settings of multiple S-BOXs to OFF except for “Picture Mode”– “Calibration” as follows.

※ The following setting is to synchronize the output of the source device input to a plurality of S -BOXs. Screen may not be uniform when not set.

Menu	2nd	3rd	Value
Picture	Picture Mode	Calibration	Calibration
	Advanced Settings	Picture Enhancer	OFF
	LED HDR	Inverse Tone Mapping	OFF
		Dynamic Peaking	OFF
		Eco Image Enhancer	OFF
		Color Mapping	OFF
	Picture Options	Color Tone	OFF
		HDMI Black Level	OFF
		Auto Motion Plus Settings	OFF
System	Power and Energy Saving	Brightness Optimization	OFF

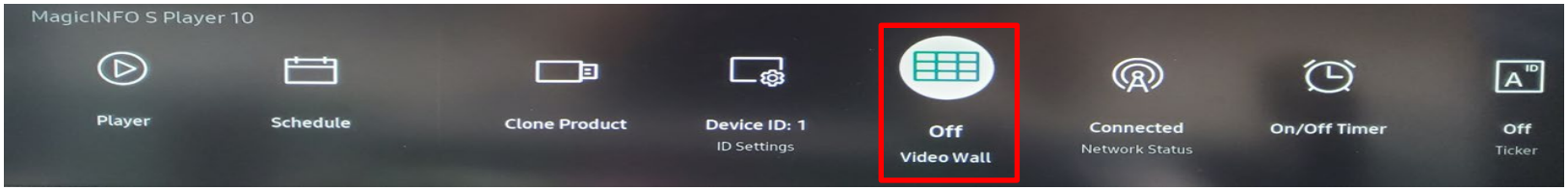


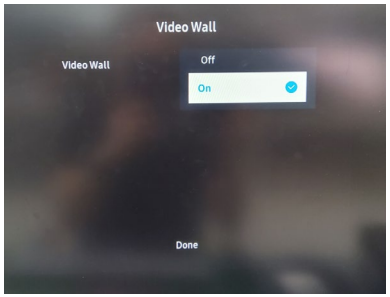


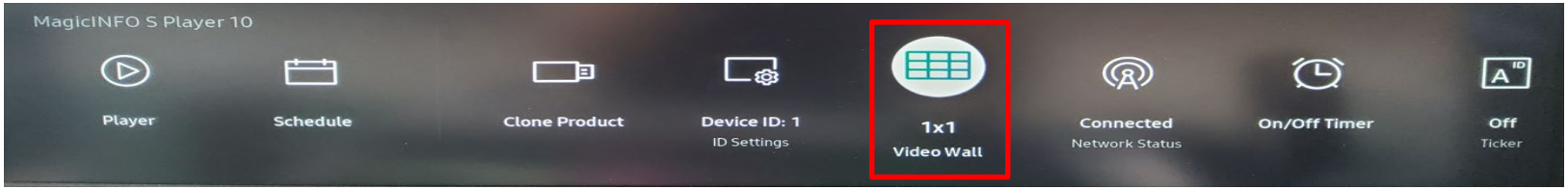
4. Setting andguide

❑ ① S-BOX Settings– Grouping setting

- Select Home → Video Wall mode using Remote Controller.
- Select Video Wall Mode as “ON”.





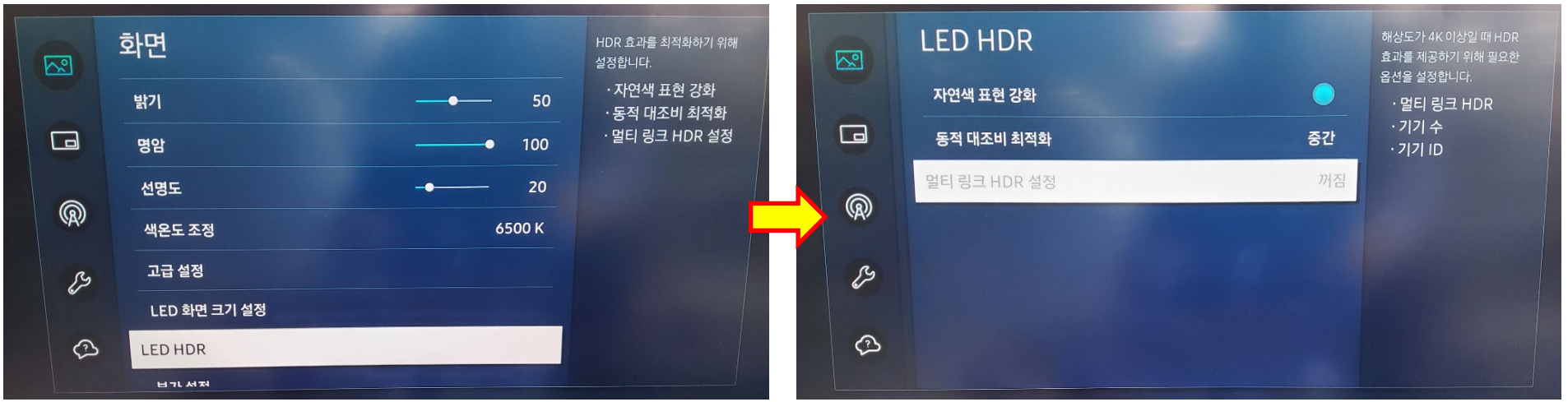


4. Setting and guide

❑ ① S-BOX Settings – Multi-Link HDR Setting

- SBB-CS4B model supports LED HDR mode.
- When configuring a screen by connecting multiple S-BOX, Multi-Link HDR Setting must be turned on. (Refer to page 17)
- Click the Menu button using the Remote Controller and select Screen → LED HDR.
- After Multi-Link HDR “On”, set the number of devices and device ID

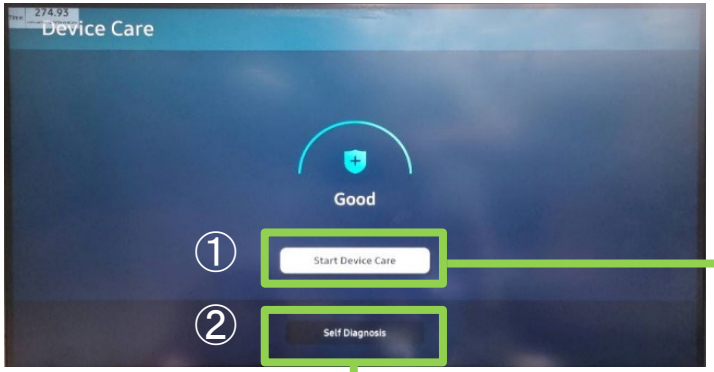
※ The maximum number of connected devices is “4”. You can set your ID to 1,2,3,4.



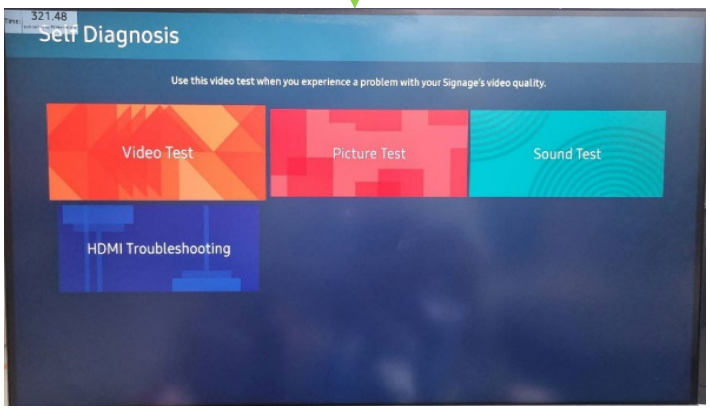
4. Guide for settings

❑ ① S-BOX– Device care

- Checkdevice condition of S-BOX and optimization can be performed with "DeviceCare“
- Menu → Support → Device Care







① Start Device CareCheck device condition of S-BOX automatically

② Self diagnosis: User can check device condition of S-BOX

1) Video Test: Test video will be shown. User can check video trouble.

2) Picture Test: Test picture will be shown. User can check picture and screen trouble.

3) Sound Test: Test sound will be played. User can check audio trouble.

(※ Audio output supported model only)

4) HDMI Troubleshooting: User can check HDMI input or cable trouble.

Adjust HDMI signal strength will be performed.

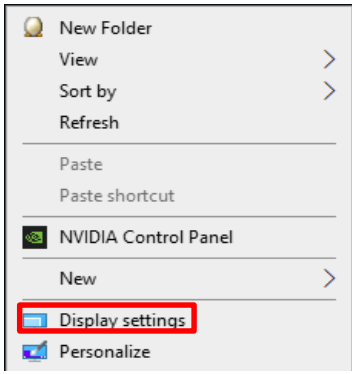
(※ Screen may blinking while adjusting HDMI signal strength. It is not a device fault.)

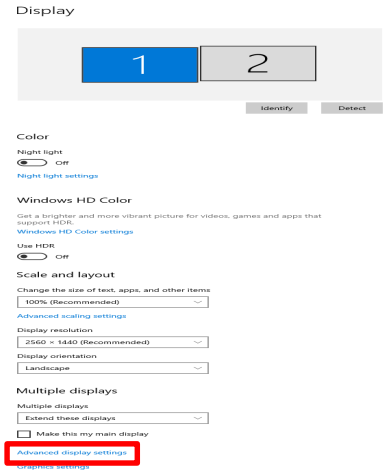
4. Setting andguide

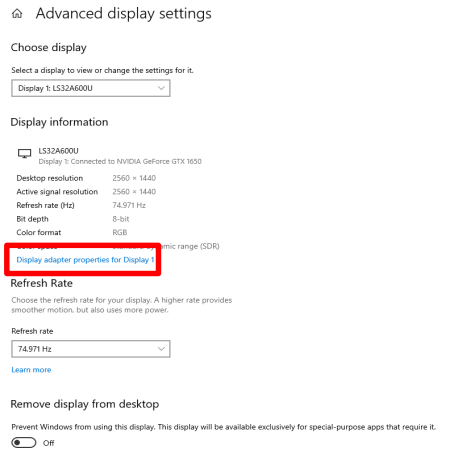
❑ ② Video Source Settings- Change Output Frequency (PC

- Change the output frequency of the PC as follows.

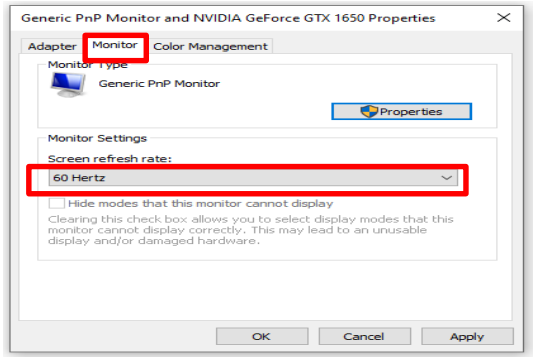
• Right-click desktop mouse → Select screen resolution → Click advanced settings → Click display adapter attribute display







• Click Monitor Tap → Monitor Settings → Change screen playback frequency to 60Hz



4. Setting andguide

❑ ② Video Source Settings- Change Output Frequency(1/3)

- Support frequency of CS4B is as follows. Change the output frequency of the video source to match the corresponding support frequency.

Resolution	Horizontal Frequency (kHz)	Vertical Frequency (Hz)	Pixel Clock (MHz)	Sync Polarity (H/V)	HDMI 1~2	DP	S-BOXGrouping S upported
IBM/VESA, 640 x 480	31.469	59.94	21.175	N / N	O	O	O
Mac, 640 x 480	35	66.667	30.24	N / N	O	O	-
VESA 640 x 480	37.861	72.809	31.5	N / N	O	O	-
VESA, 640 x 480	37.5	75	31.5	N / N	O	O	-
IBM, 720 x 400	31.469	70.087	28.322	N / P	O	O	-
VESA, 800 x 600	37.879	60.317	40	P / P	O	O	-
VESA, 800 x 600	48.077	72.188	50	P / P	O	O	-
VESA, 800 x 600	46.875	75	49.5	P / P	O	O	-
Mac, 832 x 624	49.726	74.551	57.284	N / N	O	O	-
VESA, 1024 x 768	48.363	60.004	65	N / N	O	O	-

※ When entering a resolution that does not support grouping, the cabinet screen may not be visible or screen noise may occur.

※ The interlaced mode does not support S-BOX grouping.

4. Setting and guide

❑ ② Video Source Settings - Change Output Frequency (2/3)

- Support frequency of CS4B is as follows. Change the output frequency of the video source to match the corresponding support frequency.

Resolution	Horizontal Frequency (kHz)	Vertical Frequency (Hz)	Pixel Clock (MHz)	Sync Polarity (H/V)	HDMI 1~2	DP	S-BOX Grouping Supported
VESA, 1024 x 768	56.476	70.069	75	N / N	O	O	-
VESA, 1024 x 768	60.023	75.029	78.75	P / P	O	O	-
VESA, 1152 x 864	67.5	75	108	P / P	O	O	-
VESA, 1280 x 720	45	60	74.25	P / P	O	O	O
VESA, 1280 x 800	49.702	59.81	83.5	N / P	O	O	-
VESA, 1280 x 1024	63.981	60.02	108	P / P	O	O	-
VESA, 1280 x 1024	79.976	75.025	135	P / P	O	O	-
VESA, 1440 x 900	55.935	59.887	106.5	N / P	O	O	-
VESA, 1600 x 900	60	60	108	P / P	O	O	O
VESA, 1680 x 1050	65.29	59.945	146.25	N / P	O	O	-
VESA, 1920 x 1080	67.5	60	148.5	P / P	O	O	O
VESA, 2560 x 1440	88.787	59.951	241.5	P / N	O	O	-
VESA, 3840 x 2160	67.5	30	297	P / P	X	O	-
VESA, 3840 x 2160	133.312	59.997	533.25	P / N	O	X	-
CEA861 VIC 3, 720x480	31.469	59.94	27	N / N	O	O	O
CEA861 VIC 4, 1280x720	45	60	74.25	P / P	O	O	O
CEA861 VIC 5, 1920x1080i	33.75	60	74.25	P / P	O	O	-
CEA861 VIC 7,720(1440)x480i	31.469	59.94	27	P / P	O	O	-
CEA861 VIC 16,1920 x 1080	67.5	60	148.5	P / P	O	O	O
CEA861 VIC 18, 720x576	31.25	50	27	N / N	O	O	O
CEA861 VIC 19, 1280x720	37.5	50	74.25	P / P	O	O	O

4. Setting andguide

❑ ② Video Source Settings - Change Output Frequency (3/3)

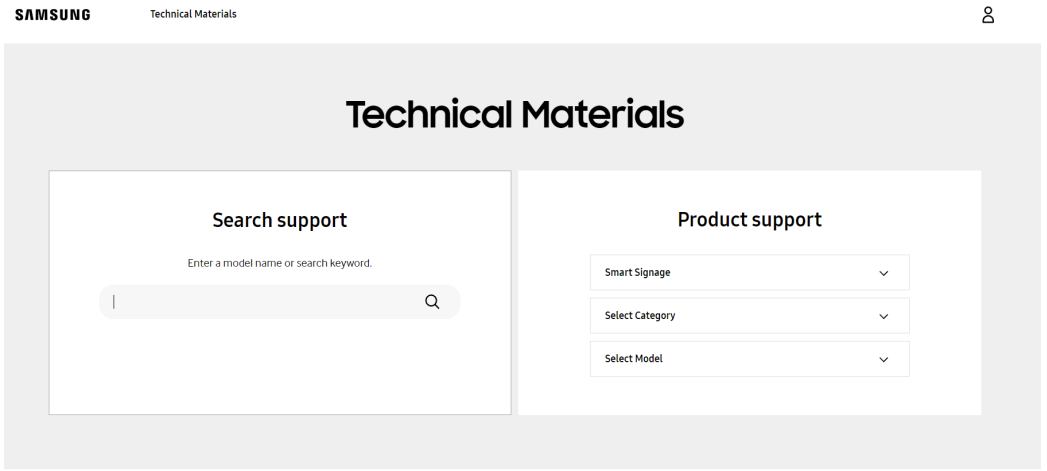
- Support frequency of CS4B is as follows. Change the output frequency of the video source to match the corresponding support frequency.

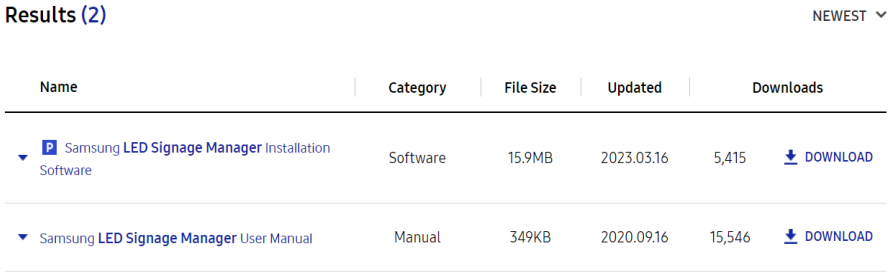
Resolution	Horizontal Frequency (kHz)	Vertical Frequency (Hz)	Pixel Clock (MHz)	Sync Polarity (H/V)	HDMI 1~2	DP	S-BOXGrouping S upported
CEA861 VIC 20, 1920x1080i	28.125	50	47.25	P / P	O	O	-
CEA861 VIC 22, 720(1440)x57(	31.25	50	27	P / P	O	O	-
CEA861 VIC 31, 1920x1080	56.25	50	148.5	P / P	O	O	O
CEA861 VIC 32, 1920x1080	27	24	74.25	P / P	O	O	-
CEA861 VIC 33, 1920x1080	28.125	25	74.25	P / P	O	O	-
CEA861 VIC 34, 1920x1080	33.75	30	74.25	P / P	O	O	-
CEA861 VIC 93, 3840x2160	54	24	297	P / P	O	O	-
CEA861 VIC 94, 3840x2160	56.25	25	297	P / P	O	O	-
CEA861 VIC 95, 3840x2160	67.5	30	297	P / P	O	O	-
CEA861 VIC 96, 3840x2160	112.5	50	594	P / P	O	X	O
CEA861 VIC 97, 3840x2160	135	60	594	P / P	O	X	O
CEA861 VIC 98, 4096x2160	54	24	297	P / P	O	O	-
CEA861 VIC 99, 4096x2160	56.25	25	297	P / P	O	O	-
CEA861 VIC 100, 4096x2160	67.5	30	297	P / P	O	O	-
CEA861 VIC 101, 4096x2160	112.5	50	594	P / P	O	X	-
CEA861 VIC 102, 4096x2160	135	60	594	P / P	O	X	-

4. Setting andguide

❑ ② Control PC Settings– LSM

- LSM: Windows Application Software for remotely adjusting the layout of LED cabinets
- LSM download path: Samsung Display Solutions (https://displaysolutions.samsung.com/support/resources/productsupport)
  Search support > Enter LED Signage Manager> Click DOWNLOAD
- ※ Partner login is required to download the program.

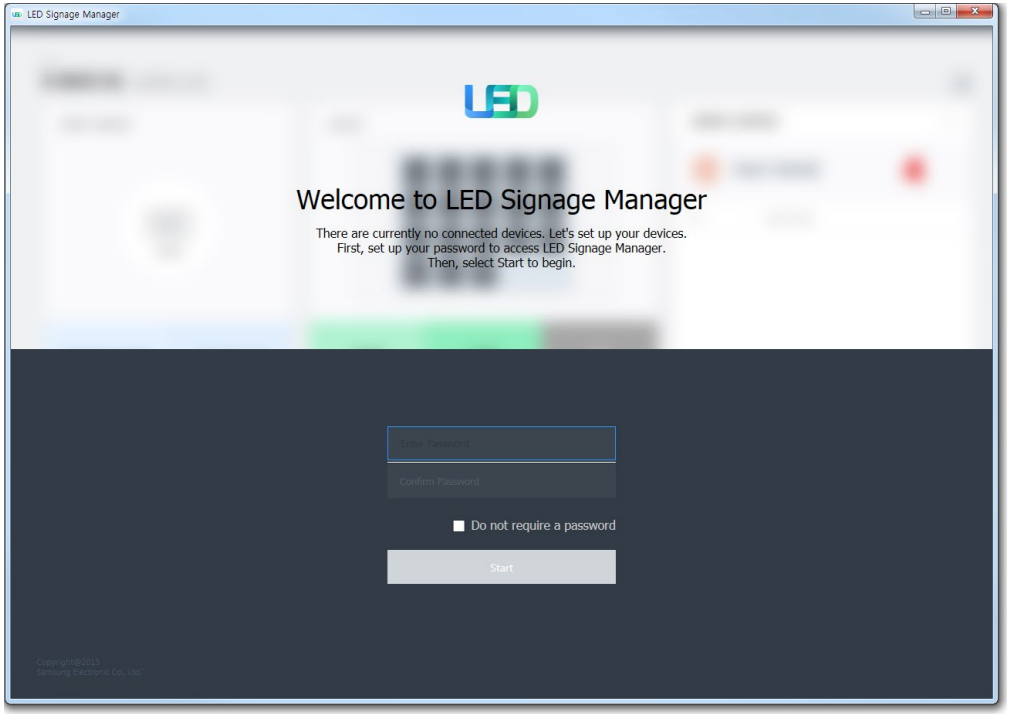




4. Setting andguide

❑ ② Control PC Settings- LSM

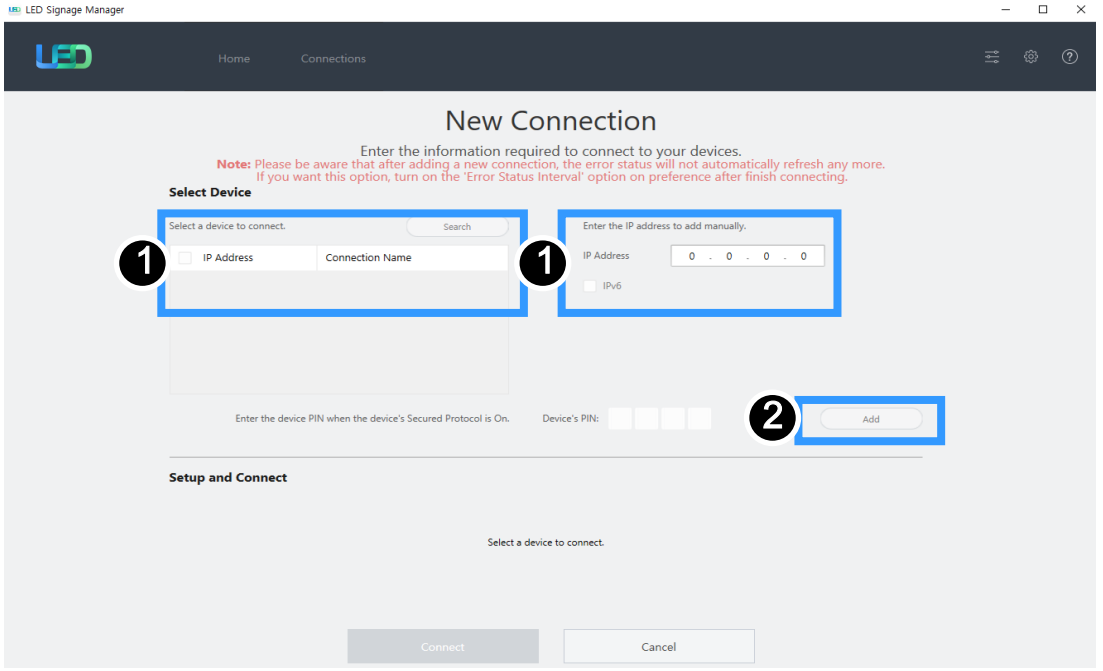
1. When LSM is executed for the first time, a page that sets the password appears.
2. To set the password, the user must enter the same password twice and click the Start button.
3. If you don't want to use password, check Don't use password option. In this case, password input is no longer required when LSM is executed.

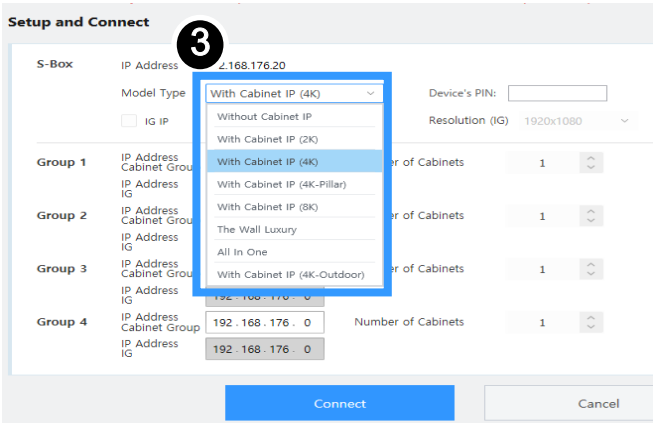


4. Setting andguide

❑ ② Control PC Settings- Connect LSM

1. To add connection information, you can enter Search or IP address directly. When you click the Search button, the IP address of the connectable S-BOX on the same network appears. If you know the IP address of S-BOX, you can also enter it directly.
2. When you click the Add button, the connection information is added to Setup and Connect.
3. The user can select the model type of S-BOX. A total of 8 model types are provided. ForSBB-CS4B connection, you can select With Cabinet IP (4K) according to the connection type.





4. Setting andguide

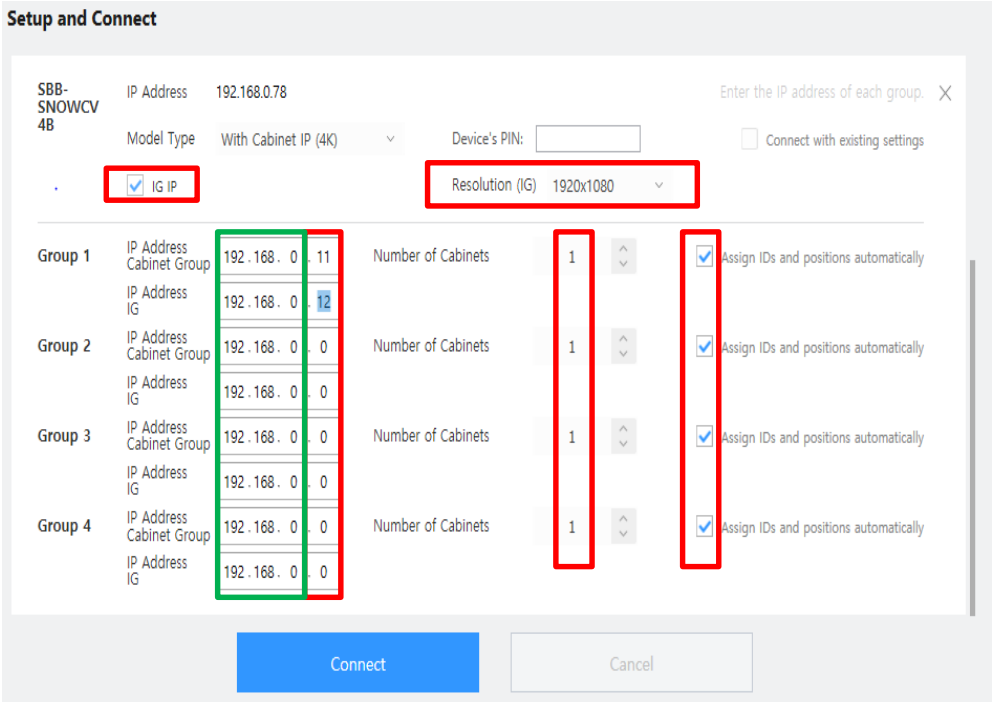
❑ ② Control PC Settings- Connect LSM

1. You can select With Cabinet IP (4K) according to the connection method of LED Cabinet. You must specify the IP address of the LED cabinets connected to each port. Sets the number of connections.
2. Check or change Resolution (IG) by checking the connected cabinet. (Refer to page 20)
3. The IP address of the IG connected to each port must be specified. Set the number of connections and click Connect.

※ If the IP is already set in the LED cabinet, you can check the “Connect with Existing Settings” option and connect quickly.

※ You can enter the IP address only in the Cabinet Group that needs actual use.

[□ Caution!]
Fixed IP settings are recommended when setting up an LED cabinet network.
If IP is changed when using DHCP, LSM connection may be disconnected.
192.168.10. the x band is used for internal communication of LED cabinet.
Please use IP except for the band.
Do not allocate IP arbitrarily, but allocate IP for LED through IT manager.

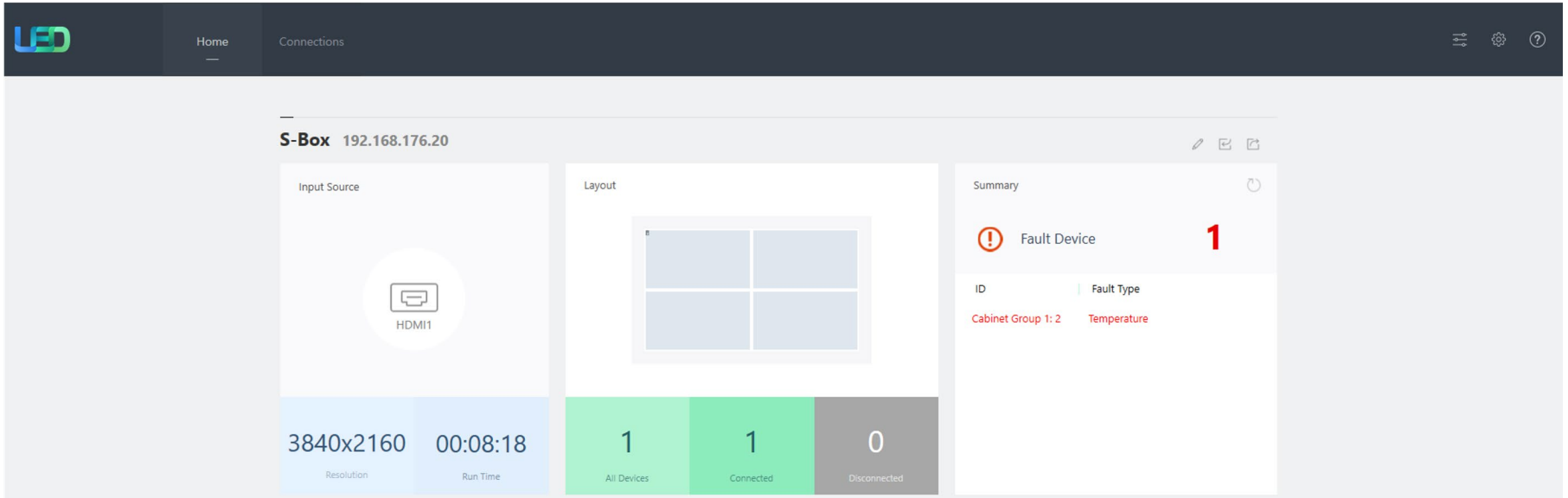


4. Setting andguide

❑ ② Control PC Settings- LSM HOME Screen

[★Caution!]
For internal communication between SBOX and LED Cabinet, use ports 1515, 48484, 48485, and 58585.
During firewall/security network use, the port must be enabled between S-BOX and LED.

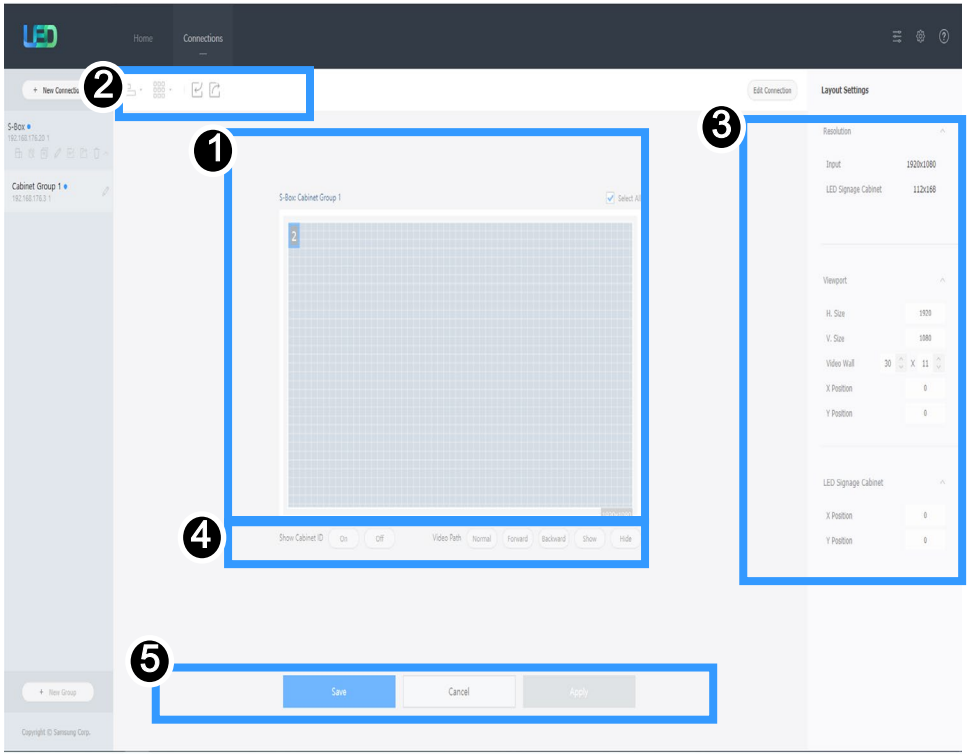
Shows the information, input source, cabinet configuration, and error devices of the devices connected through the home screen



4. Setting andguide

❑ ② Control PC Settings- Edit Connection

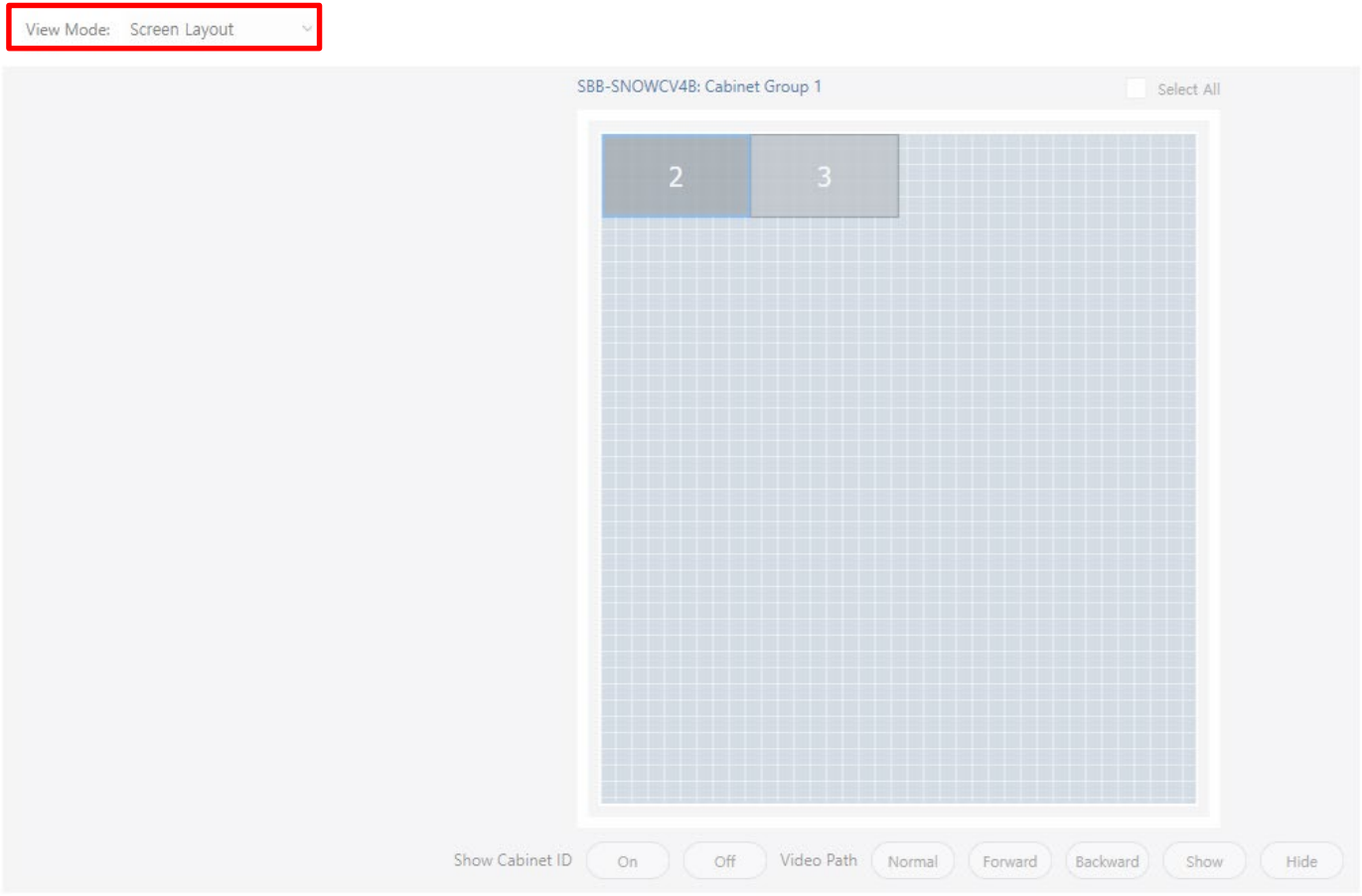
1. Connection layout: Adjust the location and layout of each LED cabinet in the output source area of the S-BOX.
2. Feature View: Provides an Edit button to modify connection information and automatic alignment of LED cabinets.
3. Device Information/Setting View:
   Shows the LED cabinet information according to the categories below. (i) Resolution: Input source resolution information (ii) ViewPort: landscape/vertical size, video wall matrix, x/y coordinate setting (iii) LED Signage Cabinet: x, y position of LED cabinet
4. Show ID: When selected, displays each ID in all connected LED cabinets. Video Path: You can check the video path setting and current FW/BW status of each cabinet of LED.
5. Press the Save button to save the status of 1 to 4.



4. Setting andguide

❑ ② Control PC Settings– Diagnostic Status

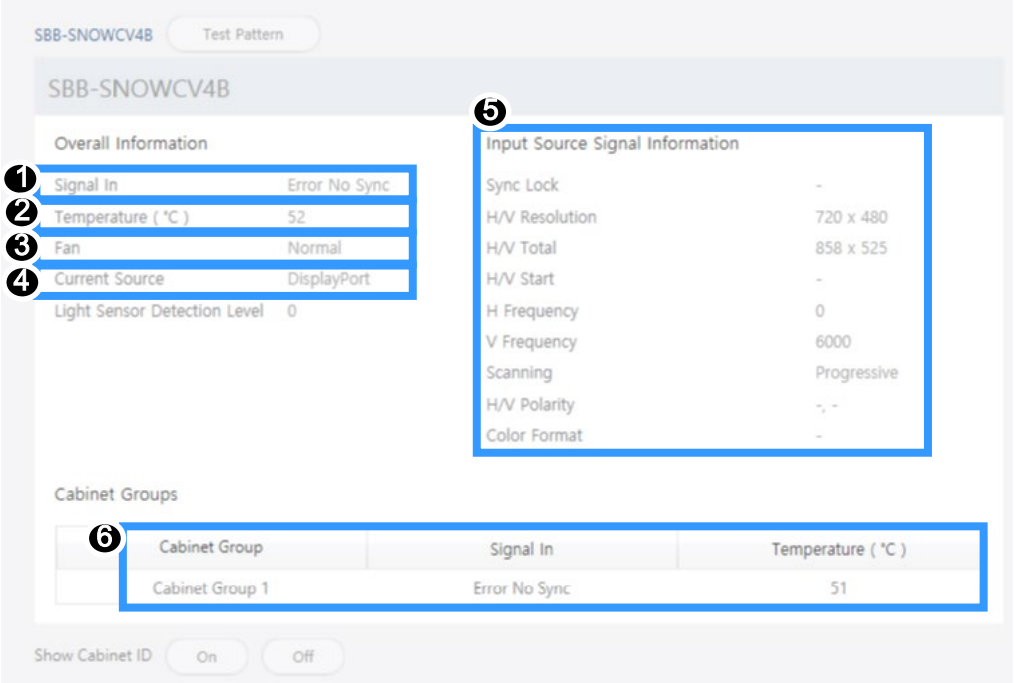
Change View Mode to “Diagnostic Status” to check the diagnostic function of LSM.



4. Setting andguide

❑ ② Control PC Settings- Diagnostic Status (SBOX)

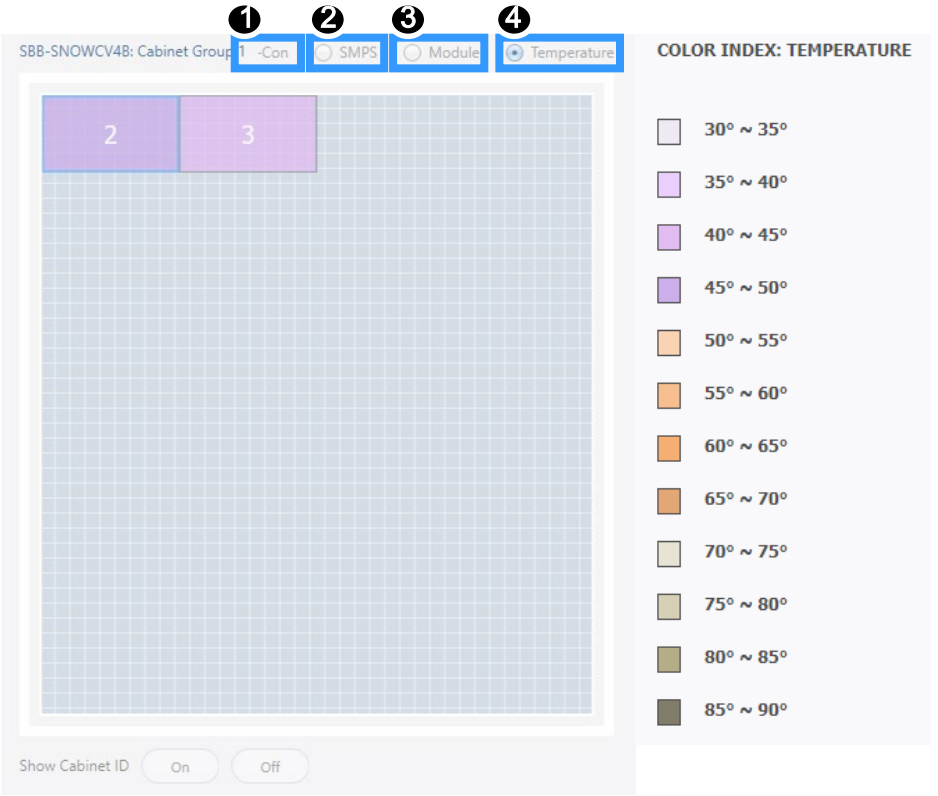
1. Signal In: Check if the input video signal is received normally.
2. Temperature (℃): Shows the current temperature of SBOX.
3. FAN: Check if Fan is operating normally. Displays “Normal” when normal and “Error” when abnormal.
4. Current Source: Displays which source the current input source is.
5. Input Source Information: Displays the video information for the input source.
6. Cabinet Groups: Displays the input signal and temperature of the connected IG.



4. Setting andguide

❑ ② Control PC Settings– Diagnostic Status (Cabinet)

1. T-CON: Shows whether the TCON of the cabinet works normally.
2. SMPS: Shows whether the SMPS of cabinet is ON/OFF.
3. Module: Shows whether the module works normally.
4. Temperature: The current temperature of the cabinet is shown in a picture. (Color differences by temperature section)



4. Setting andguide

❑ ② Control PC Settings- Device Information/Setting View

1. Basic :

   . On / Off, change input source, Screen Mute / Freeze

2. Picture

   . Picture Mode change, Brightness / Contrast / Sharpness, Color, Tint(G/R), Color Temp(K),

   Gamma, White Balance adjustment, etc.

3. Picture Options

   . Color Tone, HDMI Black Level

4. Advanced Settings

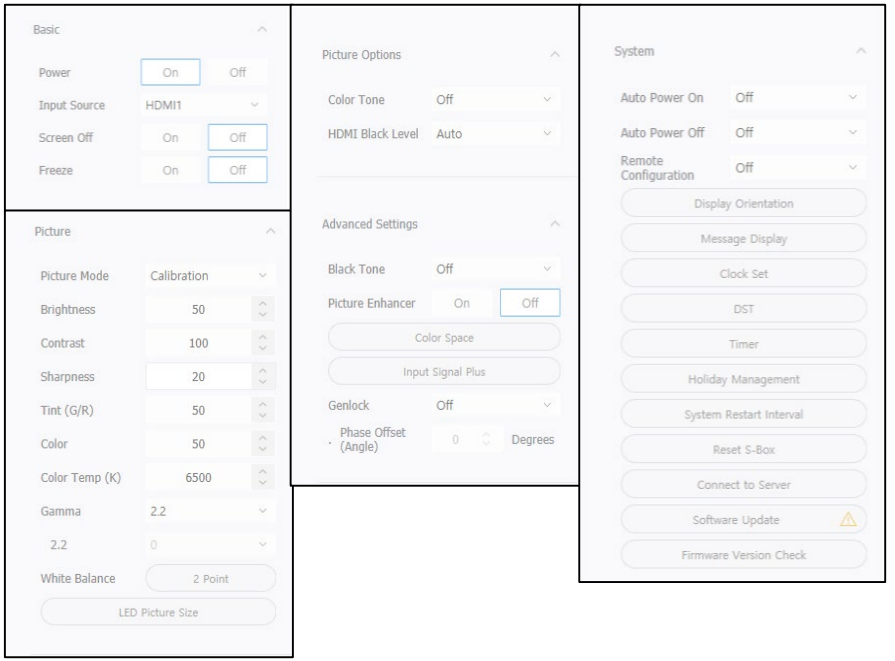
   . Black Tone, Picture Enhancer, Color Space, Input Signal Plus

5. System

   . Auto Power On/Off, Standby Control

   Clock, Timer, System Restart Interval

   Software Update Function

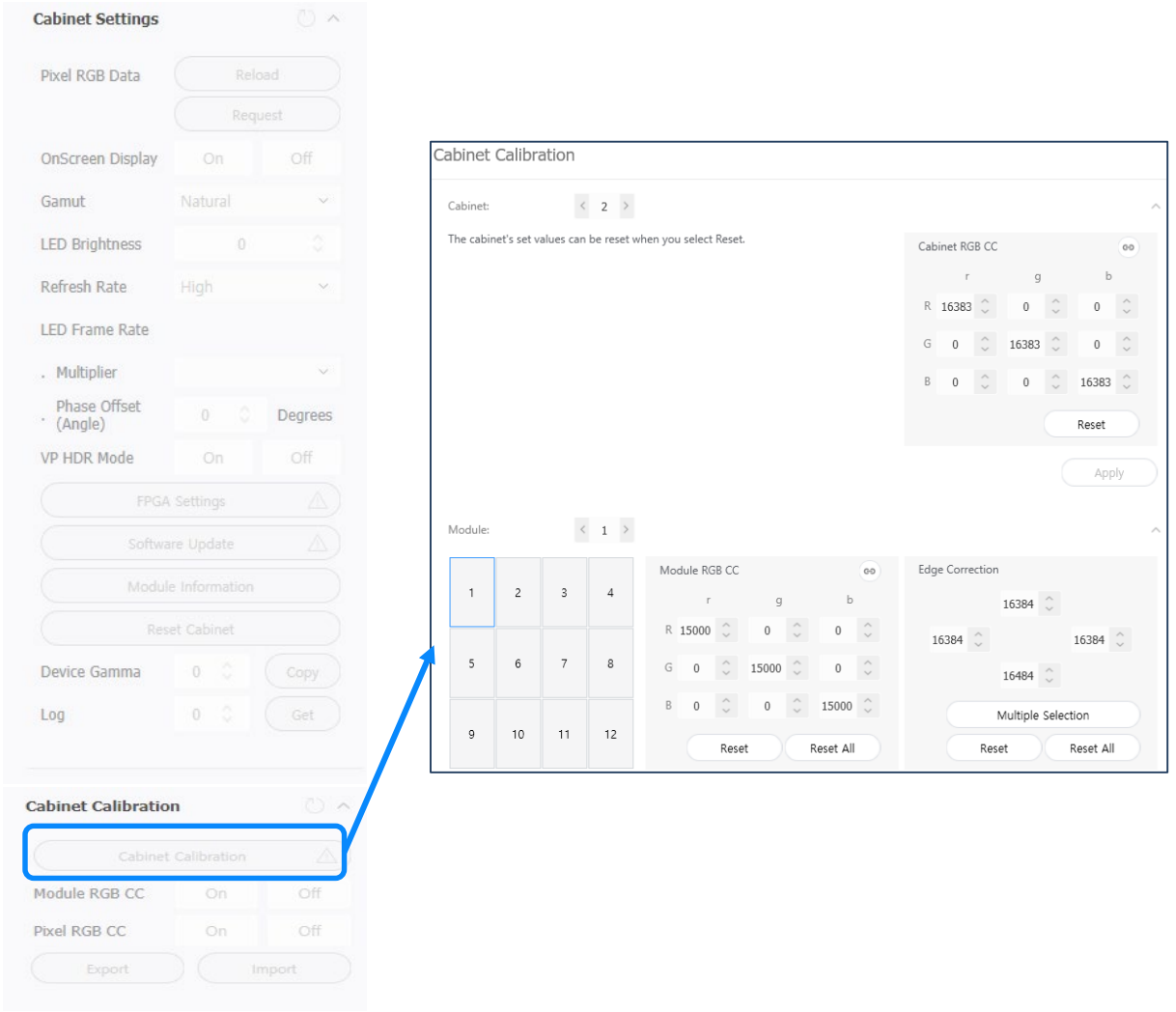


4. Setting andguide

❑ ② Control PC Settings- Device Information/Setting View

6. Cabinet Settings
   . Pixel RGB Data Reload / Request
   . OnScreenDisplay On / Off
   . Adjust LED Brightness
   . VP HDR Mode On / Off
   . Software Update function

7. Cabinet Calibration
   . Cabinet / Module / Edge adjustment
   . Gradation adjustment



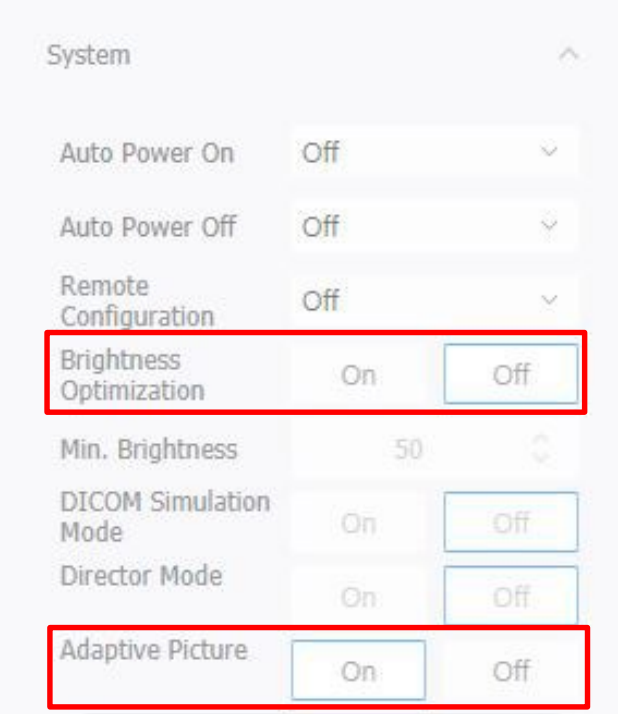
4. Setting andguide

❑ ② Control PC Settings- Eco Sensor Setting

8. Light Sensor Setting

- On Brightness Optimization for Illuminance Sensor On or On Adaptive Picture

* Both can't be turned on.



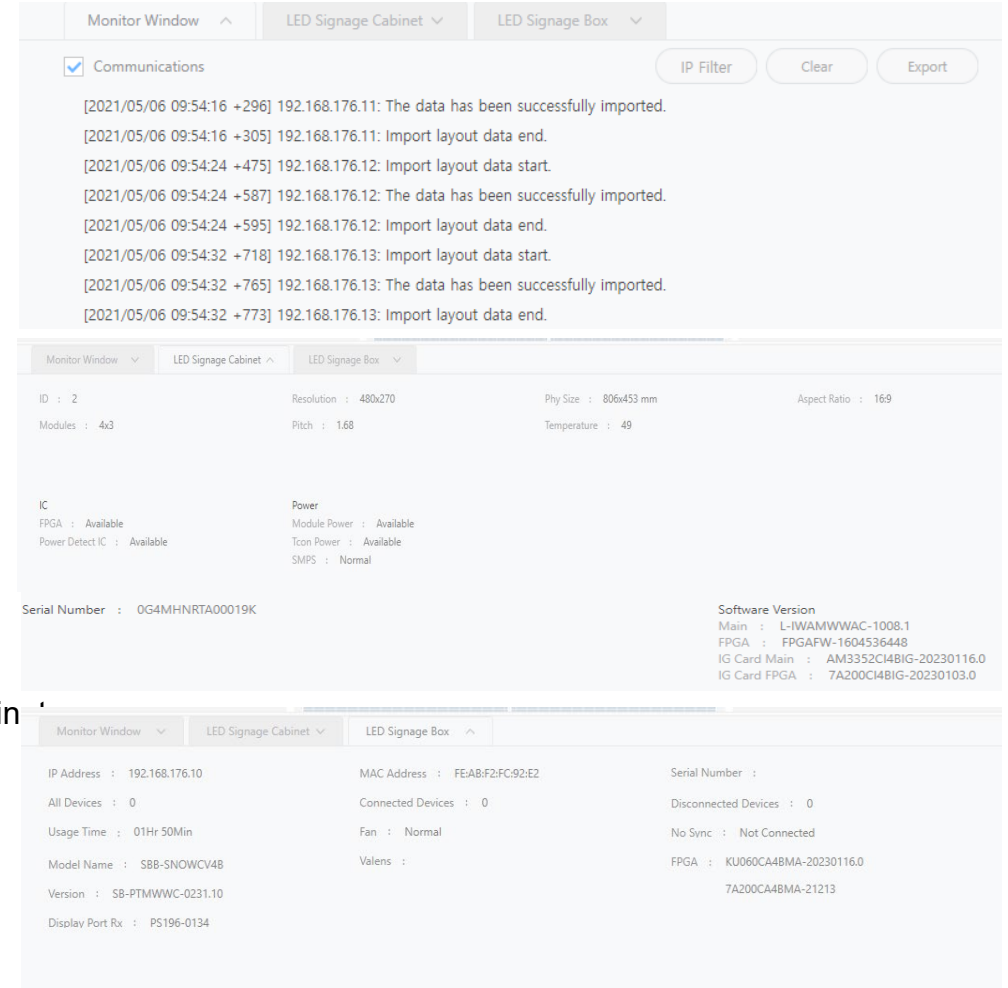
4. Setting andguide

❑ ② Control PC Settings- Sub Information View

1. Monitor Window:
   MDC communication log and connection device information can be checked
   Extractable as a file

2. LED Signage Cabinet:
   IC information and power information of LED cabinet
   F/W information of IG

3. LED Signage Box:
   IP address, MAC address, ID range of LED cabin
   Number of LED cabinets (full / connected / unconnected),
   Serial number, version information



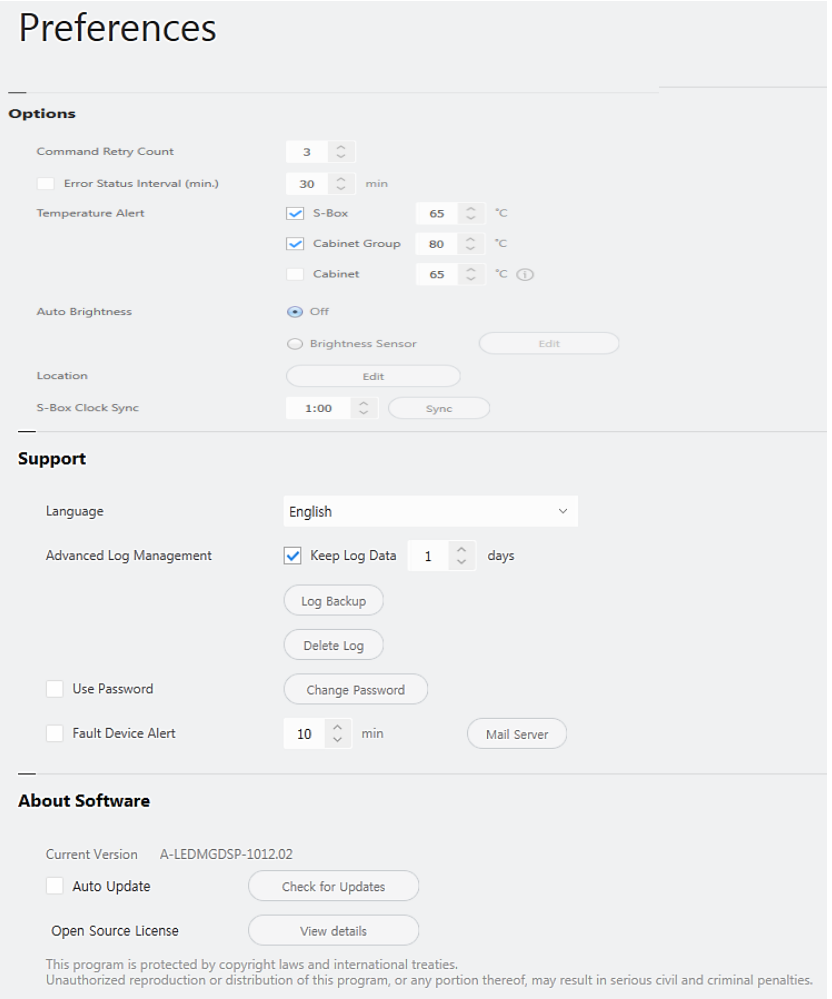
4. Setting andguide

❑ ② Control PC Settings- Preference

1. Options
   - Command retry count
   - Error status check interval
   - Temperature warning notification
   - S-Box Clock sync

2. Support
   - Program language
   - Log data management
   - Notification of device error through mail
   - Password Configuration Options

3. About Software
   - Current version and update function of LSM

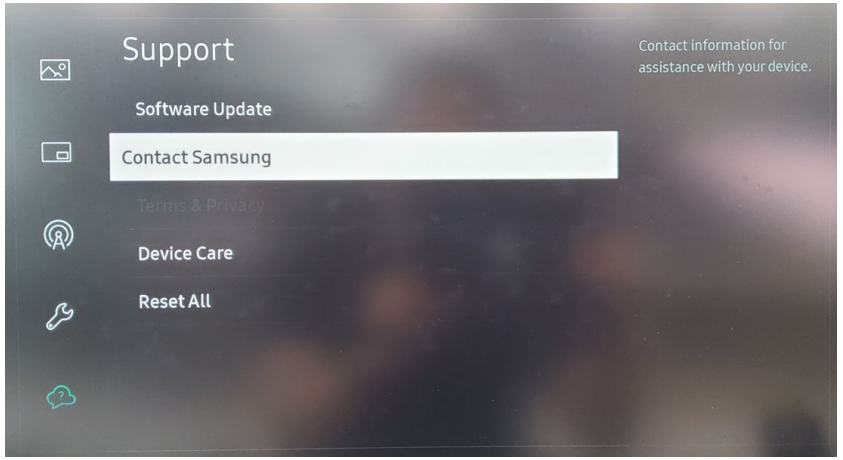


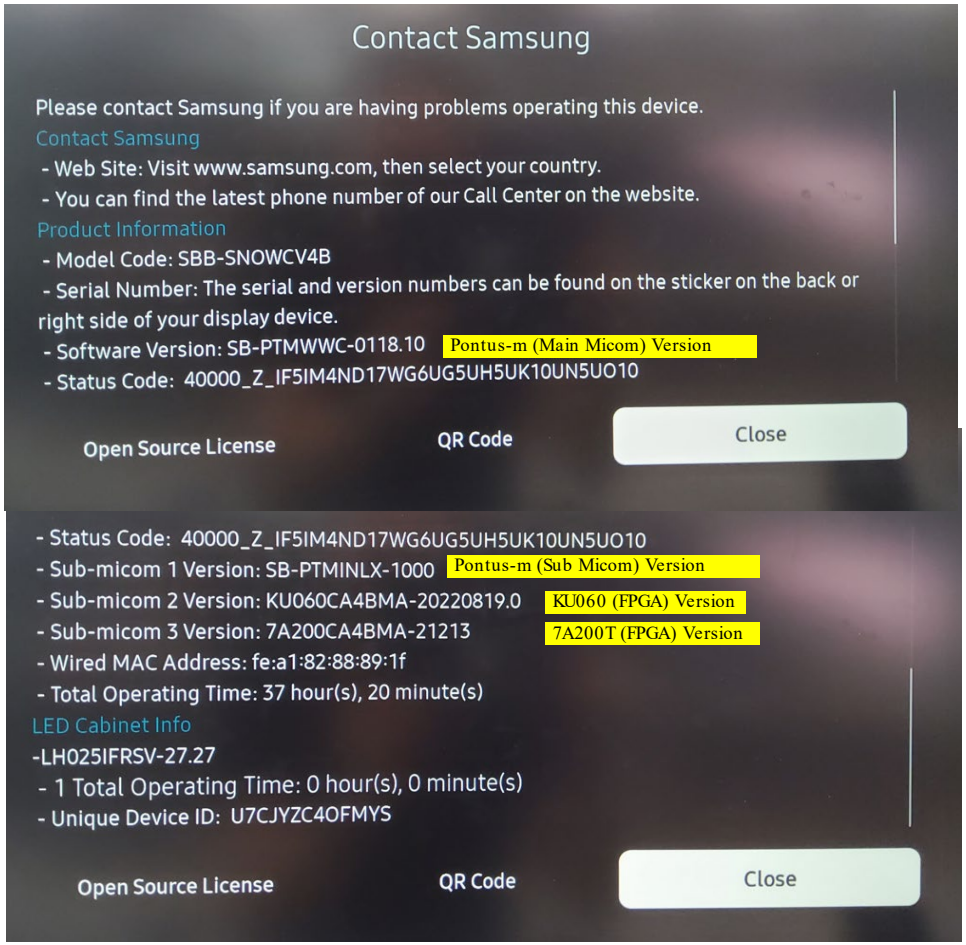
5. SW Update

5. SW Update

❑ Software Update- Check software update version

※ Contact Samsung Entry: MENU -> Support -> Contact Samsung





5. SW Update

❑ Software Update– Update using USB

① Pontus-M Main Micom Software Upgrade

- **USB Download**

" https://displaysolutions.samsung.com/support/resources/product-support " Upgrade firmware downloaded from the website

1. Connect the USB drive where the file is stored to the set.
(Please be careful not to disconnect the power or USB drive during the upgrade.)

2. When the firmware upgrade is completed, the set is automatically turned off and then turned on again.

3. Please check the firmware version after completing the upgrade.
(The new version has a higher number than the old version.)

※ Note) Since it is recognized as a file name, it is okay to have other model upgrade files together.

※ Note) After upgrading the software, the setting value specified by the existing user returns to the default value.
It is recommended to record the current setting value to facilitate reset after upgrade.

5. SW Update

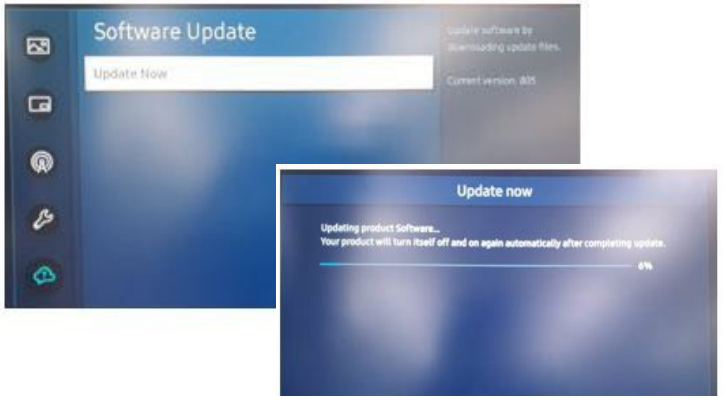
❑ Software Update– Update using USB

① Main SWUpgrade

1. Save the “SBPTMWWC” folder to the USB root.
2. Connect the USB.
3. Click the “MENU” key on the remote control.
4. Select the “Customer Support” menu.
5. Select the “Update Software” menu.
6. Select the “Update Now” menu to start the upgrade.
7. Wait for the upgrade to complete.
8. Please check SW version.



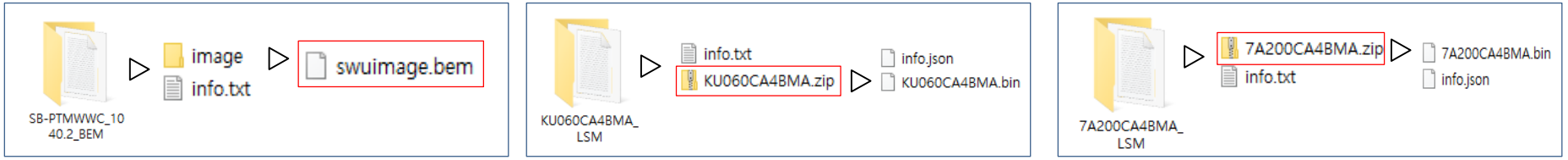




5. SW Update

❑ Software Update – Update Using LSM (S-BOX)

1. Preparethe update SW File. (Files selected when updating files marked in red)



2. After connecting LSM andCS4BPGS,Sclick the Software Update button under the S-Box Settings > System menu

3. Press the Browse button to select the prepared SW file

4. After selecting the S-Box to update, click the Update button to start the update.

❖ If the following occurs after selecting the update file, please check each content and take action..

Update Status

The software version of the device is later than the software version of the update file. Check the software version of the update file.

The version of the selected file is lower or equal to the Software version of the current S-Box.
Please prepare a higher version of the file.

Update Status

The software version names of the device and update file do not match. Check the software version name of the update file.

The version name (version TAG) of the selected file does not match the current software version name of the S-Box.
Please prepare the correct file.

Update Status

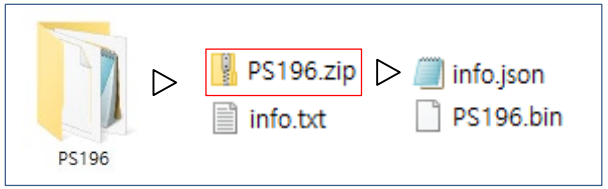
There is no info.txt file in the designated location. Check the location of the info.txt file.

the info.txt file does not exist.
Please check the file.

5. SW Update

❑ Software Update – Update Using LSM (S-BOX)

1. Preparethe update SW File. (Files selected when updating files marked in red)



2. After connecting LSM andCS4BPGS,Sclick the Software Update button under the S-Box Settings > System menu

3. Pressthe ' Browse button to select the prepared SW file.

4. After selecting the S-Box to update, click the Update button to start the update.

❖ If the following occurs after selecting the update file, please check each content and take action..

Update Status

The software version of the device is later than the software version of the update file. Check the software version of the update file.

The version of the selected file is lower or equal to the Software version of the current S-Box.
Please prepare a higher version of the file.

Update Status

The software version names of the device and update file do not match. Check the software version name of the update file.

The version name (version TAG) of the selected file does not match the current software version name of the S-Box.
Please prepare the correct file.

Update Status

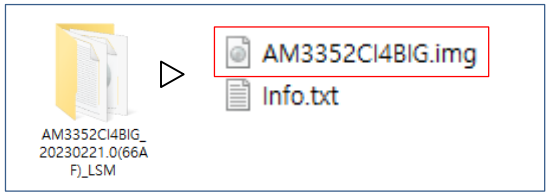
There is no info.txt file in the designated location. Check the location of the info.txt file.

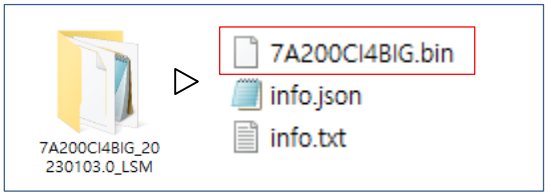
the info.txt file does not exist.
Please check the file.

5. SW Update

❑ Software Update- Update Using LSM(G)

1. Preparethe update SW File. (Files selected when updating files marked in red)





2. After connecting LSM andCS4BPGS,Sclick the Software Update button under the Cabinet Settings >System menu

3. Pressthe Browse button to select the prepared SW file.

4. After selecting the S-Box to update, click the Update button to start the update.

❖ If the following occurs after selecting the update file, please check each content and take action..

Update Status

The software version of the device is later than the software version of the update file. Check the software version of the update file.

The version of the selected file is lower or equal to the Software version of the current S-Box.
Please prepare a higher version of the file.

Update Status

The software version names of the device and update file do not match. Check the software version name of the update file.

The version name (version TAG) of the selected file does not match the current software version name of the S-Box.
Please prepare the correct file.

Update Status

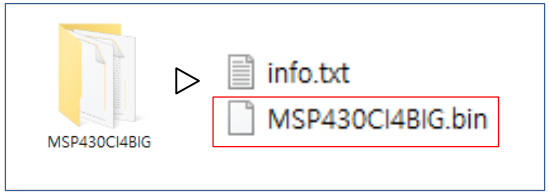
There is no info.txt file in the designated location. Check the location of the info.txt file.

the info.txt file does not exist.
Please check the file.

5. SW Update

❑ Software Update - Update Using LSM(IG)

1. Preparethe update SW File. (Files selected when updating files marked in red)



2. After connecting LSM andCS4BPGS,Sclick the Software Update button under the Cabinet Settings >System menu

3. Pressthe Browse button to select the prepared SW file.

4. After selecting the S-Box to update, click the Update button to start the update.

❖ If the following occurs after selecting the update file, please check each content and take action..

Update Status

The software version of the device is later than the software version of the update file. Check the software version of the update file.

The version of the selected file is lower or equal to the Software version of the current S-Box.
Please prepare a higher version of the file.

Update Status

The software version names of the device and update file do not match. Check the software version name of the update file.

The version name (version TAG) of the selected file does not match the current software version name of the S-Box.
Please prepare the correct file.

Update Status

There is no info.txt file in the designated location. Check the location of the info.txt file.

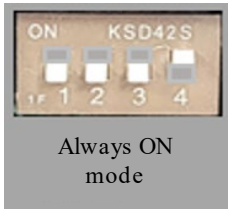
the info.txt file does not exist.
Please check the file.

Appendix

❑ IG POWER CASE

[CASE 1] Always ON mode (IG 4 deep switch ON)

※ Recommended installation specifications

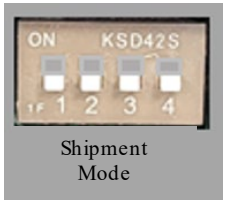


Always ON mode

S-BOX/IG POWER OFF → Cabinet Group screen OFF → S-BOX POWER ON → Cabinet Group screen ON

- In case of CASE 1, IG enters Always on state regardless of S-BOX signal.

[CASE 2] Standby mode (IG 4 deep switch OFF)



Shipment Mode

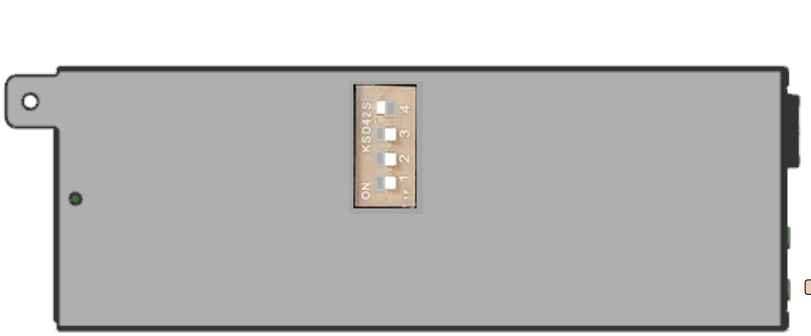
S-BOX/IG POWER OFF → Cabinet Group screen OFF → S-BOX POWER ON → Cabinet Group screen ON

- For CASE 2, if there is no S-BOX signal for more than 5 minutes, IG enters standby mode.
- To turn on the cabinet group, there is a way to connect IR to the cabinet AC power off/on or each IG. (Refer to p81, 82)

Appendix

❑ IG POWER CASE – [CASE 1]

- In the case of CASE 1, since the power of the IG is always on, there is no need to connect the IR for the front to turn on the IG (Forward).
- You can use IR for one front to control the OSD menu.

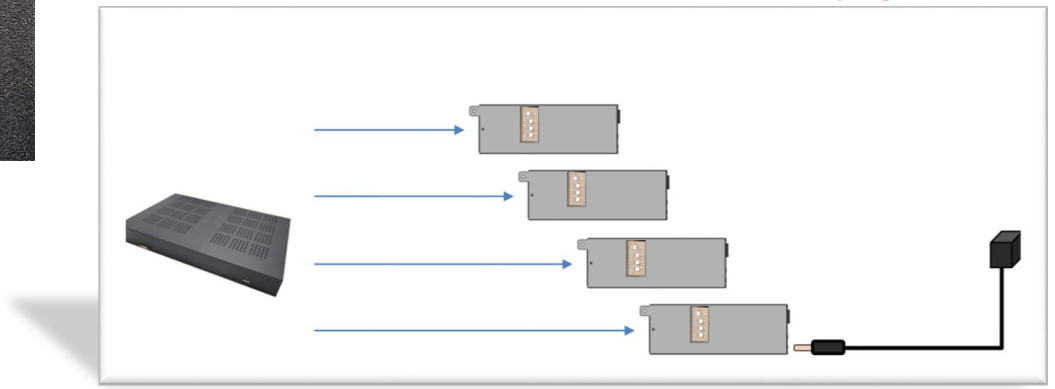


BN39-01899A

(※ In case of front IR, Additional purchase is required)



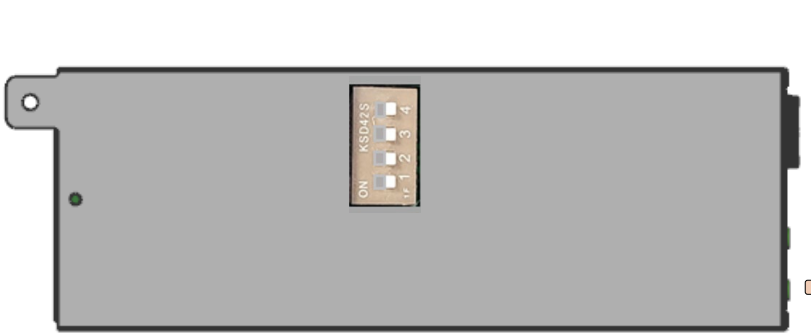
※ Connect IG and Front IR regardless of S-BOX group (1,2,3,4)



Appendix

❑ IG POWER CASE – [CASE 2]

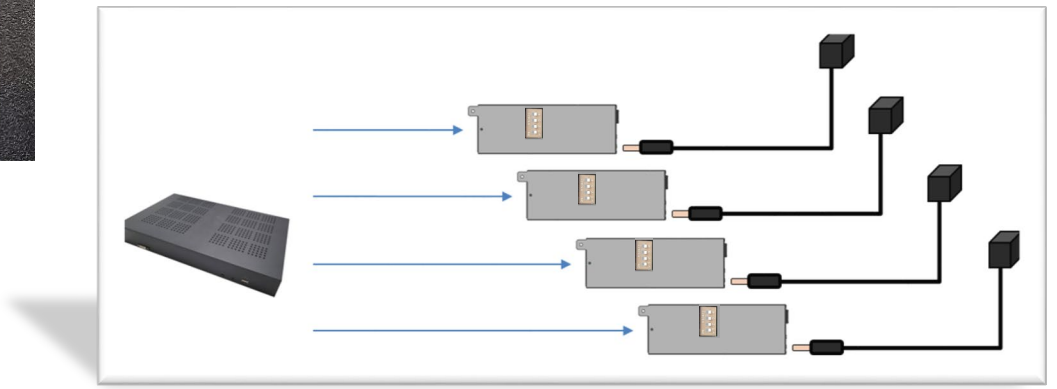
- In case of CASE 2, AC power of Cabinet must be turned off ON to turn on power of IG.
- If the above method is difficult, you can apply power by connecting IR for Front to the installed SBOX and each IG (Forward). (Note: See p41)



BN39-01899A

(※ In case of front IR, Additional purchase is required)

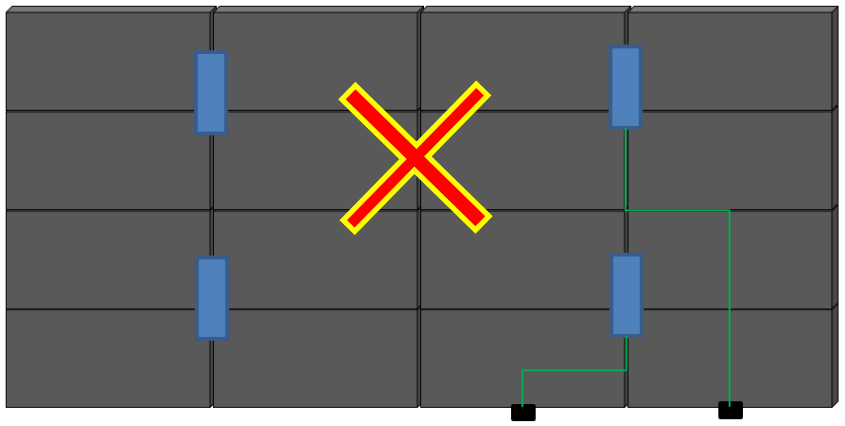


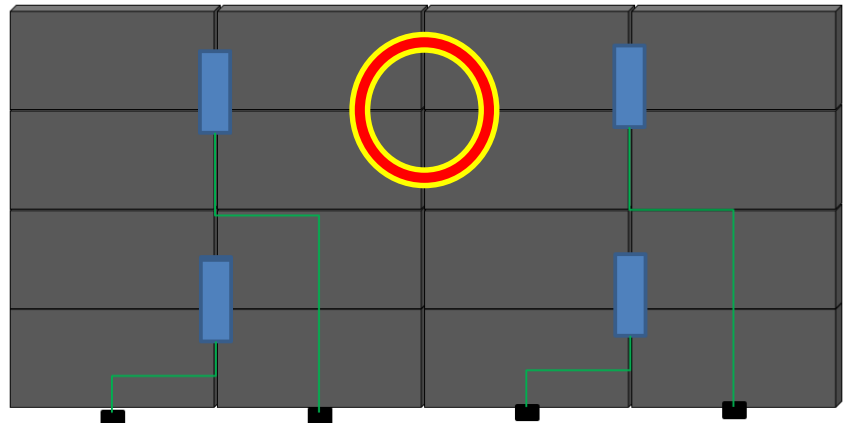


Appendix

❑ IG POWER CASE – [CASE 2]

※ For front IR installation, the screen is not turned on, the cabinet group where IR is not installed, so be careful.







Thank you

End of Document

Other ManualsLib Projects

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